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## Global DSGE models ☆

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## ABSTRACT

We introduce our GDSGE framework and a novel global solution method, called *simultaneous transition and policy function iterations (STPFIs)*, for solving dynamic stochastic general equilibrium models. The framework encompasses many well-known incomplete markets models with highly nonlinear dynamics such as models of financial crises and models with rare disasters including the current COVID-19 pandemic. Using *consistency equations*, our method is most effective at solving models featuring endogenous state variables with implicit laws of motion such as wealth or consumption shares. Finally, we incorporate this method in an automated and publicly available toolbox that solves many important models in the aforementioned topics, and in many cases, more efficiently and/or accurately than their original algorithms.

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## 1. Introduction

Dynamic stochastic general equilibrium (DSGE) models are an important tool in the study of business cycles and monetary and fiscal policies. The introduction of a general framework and local solution methods in Blanchard and Kahn (1980), Uhlig (1999), and Sims (2002) among others, and the toolbox Dynare, which implements these local methods, have made it easy to solve and estimate DSGE models and have enabled a large number of important academic studies and policy applications. However, recent developments in macroeconomics highlight the importance of solving these models using global methods. These developments include studies on

☆ First version: 02/15/2020. An online compiler server of the toolbox is deployed on the toolbox's website: <http://www.gdsge.com>, and the whole source code (including the compiler, C++ libraries) is available at <https://github.com/gdsge/gdsge>. For useful comments and discussions, we thank Jinhui Bai, Martin Evans, Luca Guerrieri, Fatih Guvenen, Mark Huggett, Tim Landvoigt, Toshi Mukoyama, John Rust, Gianluca Violante, Pengfei Wang, Ivan Wérning, and generations of students in the Advanced Macro classes at Georgetown University and Tsinghua University. We also benefited greatly from the comments and suggestions of the editor, Matteo Iacoviello, and two anonymous referees. Dan thanks Shanghai University of Finance and Economics for hospitality during the completion of the paper and Georgetown Center for Economics Research (GCER) for financial support. Wenlan Luo acknowledges the research funding supported by Tsinghua University Initiative Scientific Research Program and the National Natural Science Foundation of China (No. 72203123). Guangyu Nie acknowledges financial support from the National Natural Science Foundation of China (No. 71803124, No. 72273080), the Shuguang Program of Shanghai Education Development Foundation and Shanghai Municipal Education Commission, and Shanghai Research Center for Data Science and Decision Technology. Cristián Cuevas provided superb research assistance.

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- financial crises and highly nonlinear dynamics of the economy during financial crises, in closed or open economies such as Mendoza (2010), Bianchi (2011), Brunnermeier and Sannikov (2014), and Cao et al. (2023);
- implications of rare disasters such as Barro (2006), and Guerrieri et al. (2022) (this last paper studies the macroeconomic implications of the COVID-19 pandemic);
- portfolio choices and their implications such as Heaton and Lucas (1996) and Guvenen (2009);
- models with occasionally binding constraints (e.g., borrowing constraints and the zero lower bound, ZLB, on monetary policy) such as Gust et al. (2017), Guerrieri and Iacoviello (2017), and Cao et al. (2023);
- international finance models with endogenous capital accumulation and/or portfolio choices such as Bocola (2016), Coeurdacier et al. (2019), and Cao et al. (2020);
- and many more.

In parallel, there has also been significant progress on global solution numerical methods starting with Coleman (1990, 1991) and Judd (1992), as surveyed recently in Maliar and Maliar (2014) and Brumm et al. (2017). However, the aforementioned DSGE models are written in very different forms and the global solution algorithms to solve them are also very diverse. Therefore, there has not been a unified framework and solution method for the global solutions of these models that can be automated for users like Dynare. This paper offers such a framework and method, accompanied by an automated Dynare-like toolbox in which users only need to write model descriptions and let the toolbox translate the model descriptions into MATLAB and C++ codes that solve the model globally.

First, we develop a general framework that encompasses many recent well-known models and their extensions. The framework consists of state variables, policy variables, and short-run equilibrium conditions, e.g., market clearing conditions and Euler equations,<sup>4</sup> that fully describe the sequential equilibrium. A crucial difference between this framework and the standard DSGE framework is that the state variables and their *global* domain need to be specified, hence we name it G(lobal) DSGE. In the framework, a recursive equilibrium is a mapping from current state variables to current policy variables (policy function) and future state variables (transition function). We develop a general algorithm to robustly and efficiently solve for these recursive equilibria. The main idea of the algorithm is to solve jointly for policy and transition functions over the iterations. Hence, we call it *Simultaneous Transition and Policy Function Iterations* (STPFI) to differentiate it from the standard Policy Function Iteration (PFI) algorithms in the existing literature (e.g., Coleman (1990, 1991) and Judd (1992)), which we discuss below.<sup>5</sup> Second, we develop a toolbox that implements the algorithm. The toolbox is similar to Dynare in that it allows users to write models using intuitive and simple scripts, despite requiring users to specify the state and policy variables and the ranges for state variables explicitly, due to the nature of global solutions.<sup>6</sup> With such a design, the toolbox can be used to solve highly non-linear models with a finite number of agents (types), of small to medium sizes, with the number of state variables ranging from a few to a few dozens (using an adaptive sparse grid discretization method to deal with the curse of dimensionality). These models cover many of those in macroeconomics, international finance and asset pricing, as reviewed above and demonstrated in the examples described below. The toolbox is also useful in solving agents' decision problems in fully heterogeneous agent models with both idiosyncratic and aggregate shocks, as long as the optimal decisions can be characterized by first order conditions.

One of the challenges associated with the implementation of our algorithm is to solve for a large number of nonlinear equations with many unknowns, including future endogenous state variables in the STPFI algorithm and unknowns that may need to satisfy inequality constraints (e.g. the Lagrangian multiplier that enters a complementary-slackness condition). The toolbox addresses this challenge by using an efficient equation solver that is able to explicitly enforce bounds of unknowns,<sup>7</sup> combined with an automatic differentiation method that is able to compute the Jacobian matrix up to machine precision with small cost.<sup>8</sup> Besides, the toolbox implements all actual computations in C++ to maximize the performance,

<sup>4</sup> We use an important result in constrained optimization that for *globally* concave optimization problems, short-run first-order conditions such as the Euler equations and complementary-slackness conditions are necessary and sufficient for optimality. Duffie et al. (1994, Proposition 3.2) provide a proof for such a result in the context of an incomplete markets DSGE model.

<sup>5</sup> In simple models, the distinction between policy and transition functions is not essential (for example, in the standard neoclassical model, the policy function for investment translates simply to the transition function for capital). However, in more complex models, in which future endogenous state variables cannot be written as explicit functions of current state variables and policy variables, the distinction becomes critical as we discuss in greater detail in Section 2 and Section 3.2.

<sup>6</sup> We provide an explicit comparison of Dynare and our syntaxes in the context of the workhorse RBC model on the toolbox's website (<http://www.gdsge.com/example/rbc/rbc.html#comparisons-with-dynare>) and a simple asset pricing model with an occasionally binding ZLB from Guerrieri and Iacoviello (2015) ([http://www.gdsge.com/example/simple\\_zlb/simple\\_zlb.html](http://www.gdsge.com/example/simple_zlb/simple_zlb.html)).

<sup>7</sup> We use a constrained dogleg algorithm developed by Bellavia et al. (2012), which is tailored for nonlinear systems with simple bound constraints imposed on unknowns. The algorithm modifies a conventional trust-region-dogleg search step (Powell, 1970) by an affine-scaling transformation (Heinkenschloss et al., 1999), so that the search always stays within the predetermined bounds of unknowns. Adopting such an algorithm avoids using general constrained optimization routines to deal with the inequality constraints, which is usually less efficient than the dogleg method for solving equations, and avoids using ad-hoc transformations to convert constrained equations into unconstrained ones (as done in Cao (2018) and Elenev et al. (2021), for example,  $x = (\sin(y))^2$  if  $0 \leq x \leq 1$ ).

<sup>8</sup> Automatic differentiation (AD) is different from the finite-difference method or analytical differentiation method that are more familiar to economists. AD exploits the fact that every computer program, no matter how complicated, executes a sequence of elementary arithmetic operations, of which the derivatives can be evaluated analytically. By applying the chain rule repeatedly to these operations, the derivative of a complex function can be computed along calculating the value of the function. Because at each step of the chain rule the derivative is evaluated analytically, AD can calculate gradients up

and provides a MATLAB interface that allows users to specify options and generate model output conveniently. This efficient implementation enables the toolbox to solve many classical models that are well-known for computational challenges with a few lines of toolbox codes and in a few minutes on a regular laptop (more details on the run-time is provided in the paper).

The toolbox, together with the STPFI algorithm, demonstrates its convenience and greatest power, relative to other solution methods, for models with endogenous state variables with implicit laws of motion, such as wealth shares or consumption shares. As we make clear in the applications, these endogenous state variables help reduce the number of state variables to be kept track of in models with multiple assets such as Heaton and Lucas (1996), Kubler and Schmedders (2003), and Cao (2018), or help simplify the feasible region of the endogenous state space in models with a collateral constraint such as Mendoza (2010) and Cao and Nie (2017). The endogenous state variables also help circumvent multiple equilibria issues as demonstrated in Cao et al. (2023). Our STPFI algorithm includes the vector of future realizations of endogenous state variables in the vector of unknowns to be solved at each collocation point over the iterations. The additional equations in the system of equations at each collocation point are the *consistency equations* that impose the future endogenous state variables to be consistent with current policy variables.

We provide many examples of existing seminal applications that can be solved easily using the toolbox. The examples in the paper include Heaton and Lucas (1996), Guvenen (2009), Mendoza (2010), Bianchi (2011), Barro et al. (2017), and a dynamic stochastic extension of Guerrieri et al. (2022). Each of the examples listed can be implemented within 200 lines of toolbox code and can be executed in a few minutes on regular laptops. In many cases, the toolbox algorithm is fundamentally different from the original solution methods and is either more efficient and faster or more accurate, or both. For example, in Guvenen (2009), his solution method is based on the algorithm in Krusell and Smith (1998) using fixed point iterations over the pricing and aggregate state transition functions with nested value function iterations. Our algorithm recognizes that because the agents' optimization problems are *globally* concave problems, the first-order conditions are sufficient for optimality (without having to solve the agents' Bellman equation). Therefore, we can directly use STPFIs to solve jointly for agents' optimization problems and market clearing conditions. As a result, our toolbox's solution method is significantly faster and more stable than the original one in Guvenen (2009). The baseline model in Guvenen (2009) can be solved in less 3 minutes on a regular laptop (MacBook Air 2.2 GHz Intel Core i7) while it takes about 9 hours 15 minutes (on a MacPro Workstation with 8-core 3.2 GHz Xeon processors) using the original algorithm in Guvenen (2009) as reported in Appendix A in his paper.

In another example, Barro et al. (2017) solve an incomplete markets model with rare disasters and heterogeneous risk-aversion using a mixture of projection and perturbation methods developed in Fernández-Villaverde and Levintal (2018), while our toolbox's algorithm is a pure projection method using wealth share as an endogenous state variable with an implicit law of motion accommodated by STPFI. As Barro et al. (2017) discuss in their paper, their solution method is not sufficiently accurate for large values of risk-aversion coefficients (up to 10). We show that our method can tackle cases with risk-aversion coefficients as high as 100 effectively and uncover new economic insights in these cases. We provide many more examples on the toolbox's website.

Since our toolbox solves explicitly for policy functions and transition functions, we can conveniently use them to carry out numerical accuracy analyses in terms of Euler equation errors along the lines in Judd (1992), Judd et al. (2011), and Guerrieri and Iacoviello (2015). To illustrate this feature, we provide the toolbox code for these analyses for our leading example from Heaton and Lucas (1996). The toolbox can also automatically generates Monte Carlo simulations. These simulations are useful to understand the transmission mechanisms in the models using impulse response functions and to eventually estimate the models. The code to generate impulse response functions from the dynamic extension of Guerrieri et al. (2022) is available on the toolbox's website.

The examples mentioned above are highly nonlinear models which require solutions with high accuracy but are of small scale (up to three continuous state variables). The policy and transition functions in these models can be well approximated using spline approximations. Cao et al. (2023) is a New Keynesian application with five continuous state variables using piece-wise linear approximation. Cubic spline or adaptive sparse grid approximation does not work well for the model because the policy functions exhibit sharp changes in slopes in regions where the zero lower bound or collateral constraint switches from non-binding to binding and vice-versa. However, in principle, our toolbox can accommodate medium scale DSGE models of up to 40 continuous state variables such as Smets and Wouters (2007) which features around 20 continuous state variables.<sup>9</sup> For these models with larger numbers of state variables, we provide the option of using the adaptive sparse grid method developed by Ma and Zabarar (2009) and introduced to economics by Brumm and Scheidegger (2017). This method works well when the nonlinearity in the models is not as extreme as in Cao et al. (2023). On the toolbox's website,

to the machine precision by using only a *constant* factor more operations than evaluating the original function, and is thus more accurate and much faster than the finite-difference method which requires perturbing the function along each dimension of the arguments with a finite step size. Moreover, because the chain rules are applied automatically, AD saves the hassle of taking analytical derivatives of a complex function, and applies to functions whose analytical derivatives are state-dependent (e.g., the max or min operator which enters a complementarity constraint). We use the AD implementation by Adept (Hogan, 2014) which further utilizes the expression template feature of C++ that significantly increases the efficiency of gradient evaluations.

<sup>9</sup> Notice that, in general, highly nonlinear models with fewer state variables could potentially be more difficult to solve than less nonlinear model with more state variables.

we provide the GDSGE codes for Cao et al. (2023) and a multi-country business cycle models with 15 countries and 30 continuous state variables.<sup>10</sup>

Our approach to solving models with endogenous state variables with implicit laws of motion using STPFI generalizes the approach in Cao (2010), Brumm et al. (2015a,b, 2023), Cao and Nie (2017), Cao (2018) and Cao et al. (2023). While these papers only use wealth shares as endogenous state variables, we provide a general formulation of the approach that applies to a broad class of models, including finance and asset pricing models that use consumption shares as endogenous state variables. We show that the approach is critical in enabling a simple and automated toolbox implementation. We also provide an explicit comparison with other approaches in the existing literature in the context of the Heaton and Lucas' model and find that our approach is also numerically superior for this model. One of the other approaches is the nested-fixed point algorithm proposed by Kubler and Schmedders (2003). The algorithm is based on the existing PFI algorithm. In each iteration, the authors solve for future endogenous state variables using the consistency equations as an additional fixed point problem. The solution to the fixed-point problem is then used to formulate a system of equations and unknowns for current policy variables. This nested-fixed point procedure is not amenable to a simple and automated toolbox implementation and might be unstable.<sup>11</sup> Another approach to tackle the implicit laws of motion is, at each policy function iteration, to use the transition function from the previous iteration and only solve for the policy function, instead of solving for both simultaneously as in our toolbox. Combining the transition function with the policy function from the previous iteration, one can obtain the forecasts of future variables. This approach is used in more recent papers such as Elenev et al. (2021). We call this method *transition function iterations* (TFI).<sup>12</sup> The advantage of our approach is that the equilibrium in each iteration corresponds to the equilibrium in a finite-horizon economy. For many models, this property guarantees the existence of solution to the short-run equilibrium systems. The convergence of the policy functions over the iterations thus reflects the convergence of equilibria in finite-horizon economies to those of the infinite-horizon economies, consistent with the theoretical proofs for equilibrium existence in infinite-horizon incomplete markets economies such as Duffie et al. (1994), Magill and Quinzii (1994), and more recently Cao (2020). The TFI approach does not always work for the examples considered in the present paper.<sup>13</sup>

An earlier attempt in providing a general, unified framework for global solutions of DSGE models is Winschel and Kratzig (2010). Our framework is more general and allows for endogenous state variables with implicit laws of motion, or equivalently, implicit state transition equations. We also provide a toolbox which only requires users to provide model files, similar to Dynare. Users do not need to code their model in specific programming languages such as Java, Fortran, or MATLAB. Winschel and Kratzig (2010)'s algorithm uses PFIs based on earlier work for DSGE economies such as Coleman (1990, 1991) and Judd (1992).

Our framework is more readily applicable to solving, globally, DSGE models with a finite number of agents or, more precisely, a finite number of agent types.<sup>14</sup> Cao (2020) shows that incomplete markets models with finite agent types are useful special cases of fully heterogeneous agent, incomplete markets model with both idiosyncratic and aggregate shocks à la Krusell and Smith (1998). In particular, the former corresponds to the latter in which idiosyncratic shocks are perfectly persistent. We provide an explicit comparison between the two models on the toolbox's website. In addition, the toolbox can be used to solve the agents' decision problem and to simulate in the latter given conjectured laws of motion of the aggregate state variables. Then, with an additional fixed-point iteration on these laws of motion, which can be coded simply in MATLAB, the toolbox solution can be used to solve for the DSGE in the latter. In the last section of the paper, we show how this idea can be used to solve Krusell and Smith's model in fewer than 200 lines of code.

The remainder of the paper is organized as follows. In Section 2, we present the leading example for our toolbox. In Section 3, we provide the general framework and algorithm. A wide range of examples is presented in Section 4. In Section 5 we discuss the application of our toolbox to heterogeneous agent models with both idiosyncratic and aggregate shocks. Section 6 concludes. The design and implementation of the toolbox are presented in the appendix and on the toolbox's website.

## 2. A leading example

We use the benchmark model in Heaton and Lucas (1996) as the first illustration of how to write models in our framework and solve them using the toolbox. We follow closely the notation in the original paper.

This is an incomplete markets model with two representative agents  $i \in \mathcal{I} = \{1, 2\}$  who trade equity shares and bonds. The aggregate state  $z \in \mathbf{Z}$ , which consists of capital income share, agents' income share, and aggregate endowment growth,

<sup>10</sup> <http://www.gdsge.com/example/ZLB/zlb.html> and [http://www.gdsge.com/example/multi\\_country\\_rbc/multi\\_country\\_rbc.html](http://www.gdsge.com/example/multi_country_rbc/multi_country_rbc.html) (and Online Appendix B).

<sup>11</sup> For example, the nested fixed point problems might not have a solution in early iterations when the initial guess for policy functions is far from the converged functions.

<sup>12</sup> Recently, Mendoza and Villalvazo (2020) show that this method yields significant speed improvements for models with explicit laws of motion, including Mendoza (2010)'s model with natural state variables.

<sup>13</sup> This is probably because the approach does not allow for feedback from future variables on current policy variables within each iteration. Some of the feedback effects can be introduced by manually setting dampening parameters. However, it is possible that for other models, our approach does not work and the alternative approach does. Therefore, in the toolbox, we allow for the option to use the alternative approach (the details are provided in Online Appendix A.1, in particular footnote 32).

<sup>14</sup> There is a continuum of price-taking agents within each type and they make identical decisions in equilibrium.

follows a first-order Markov process.  $p_t^s(z^t)$  and  $p_t^b(z^t)$  denote share price and bond price at time  $t$  and in shock history  $z^t = \{z_0, z_1, \dots, z_t\}$ . To simplify the notation, we omit the explicit dependence on the shock history, e.g.,  $p_t^s$  stands for  $p_t^s(z^t)$ .

Agent  $i$  takes the share and bond prices as given and maximizes her inter-temporal expected utility

$$\mathcal{U}_t^i = \mathbb{E}_t \left[ \sum_{\tau=0}^{\infty} \beta^\tau \frac{(c_{t+\tau}^i)^{1-\gamma}}{1-\gamma} \right]$$

subject to

$$c_t^i + p_t^s s_{t+1}^i + p_t^b b_{t+1}^i \leq (p_t^s + d_t) s_t^i + b_t^i + Y_t^i$$

and

$$s_{t+1}^i \geq 0, b_{t+1}^i \geq K_t^b,$$

where  $Y_t^a$  denotes the aggregate income.  $d_t = \delta_t Y_t^a$  is total dividend (capital income) and  $Y_t^i = \eta_t^i Y_t^a$  is labor income of agent  $i$ . Aggregate income grows at a stochastic rate  $\gamma_t^a = \frac{Y_t^a}{Y_{t-1}^a}$ .  $z_t = \{\gamma_t^a, \delta_t, \eta_t^1\}$  follows a first-order Markov process estimated using U.S. data. The borrowing limit is set to be a constant fraction of per capita income, i.e.,  $K_t^b = \bar{K}^b Y_t^a$ .

In equilibrium, prices are determined such that markets clear in each shock history:

$$s_t^1 + s_t^2 = 1, \text{ and } b_t^1 + b_t^2 = 0.$$

As in Kubler and Schmedders (2003), Brunnermeier and Sannikov (2014), and Cao (2018), we use the normalized financial wealth share

$$\omega_t^i = \frac{(p_t^s + d_t) s_t^i + b_t^i}{p_t^s + d_t}$$

as an endogenous state variable. In equilibrium, the market clearing conditions imply that  $\omega_t^1 + \omega_t^2 = 1$ .

For any variable  $x_t$ , let  $\hat{x}_t$  denote the normalized variable:  $\hat{x}_t = \frac{x_t}{Y_t^a}$  (except  $b_t^i$  for which  $\hat{b}_t^i = \frac{b_t^i}{Y_{t-1}^a}$ ). Using this normalization, agent  $i$ 's budget constraint can be rewritten as

$$\hat{c}_t^i + \hat{p}_t^s s_{t+1}^i + p_t^b \hat{b}_{t+1}^i \leq (\hat{p}_t^s + \hat{d}_t) \omega_t^i + \hat{Y}_t^i.$$

The wealth share is rewritten as

$$\omega_t^i = \frac{(\hat{p}_t^s + \hat{d}_t) s_t^i + \frac{\hat{b}_t^i}{\gamma_t^a}}{\hat{p}_t^s + \hat{d}_t}.$$

The optimality of agent  $i$ 's consumption and asset choices is captured by first-order conditions in  $s_{t+1}^i$  and  $b_{t+1}^i$ :

$$1 = \beta \mathbb{E}_t \left[ \left( \frac{\hat{c}_{t+1}^i}{\hat{c}_t^i} \right)^{-\gamma} (\gamma_{t+1}^a)^{1-\gamma} \frac{\hat{p}_{t+1}^s + \hat{d}_{t+1}}{\hat{p}_t^s} \right] + \hat{\mu}_t^{i,s},$$

$$1 = \beta \mathbb{E}_t \left[ \left( \frac{\hat{c}_{t+1}^i}{\hat{c}_t^i} \right)^{-\gamma} (\gamma_{t+1}^a)^{-\gamma} \frac{1}{p_t^b} \right] + \hat{\mu}_t^{i,b},$$

where  $\hat{\mu}_t^{i,s} \geq 0$  and  $\hat{\mu}_t^{i,b} \geq 0$  are the Lagrangian multipliers on agent  $i$ 's short-sale constraint and borrowing constraint, respectively. The multipliers and portfolio choices satisfy the complementary-slackness conditions:

$$0 = \hat{\mu}_t^{i,s} s_{t+1}^i,$$

$$0 = \hat{\mu}_t^{i,b} (\hat{b}_{t+1}^i - \bar{K}^b).$$

Because the optimization problems of the agents are *globally* concave, the first-order conditions are necessary and sufficient for optimality.

We solve the model using simultaneous transition and policy function iterations that we describe in more detail in Section 3: In each iteration, we take the policy functions solved from the last iteration as given, and look for pricing, allocation, and Lagrange multipliers as functions of wealth shares and exogenous states that satisfy the market clearing conditions and first-order conditions. The main challenge of doing so is immediately clear because to evaluate future prices and quantities, which show up in the first-order conditions, one needs to know the transition of the endogenous state,



$\omega_{t+1}^1$  in the current example. However,  $\omega_{t+1}^1$  depends on future prices and allocations which are themselves functions of the endogenous state variable, or in other words, the law of motion for the endogenous state variable is implicit. In dealing with such an endogenous state variable featuring an implicit law of motion, the literature has developed different methods such as the nested fixed point algorithm (Kubler and Schmedders, 2003) and the transition function iteration algorithm (Elenev et al., 2021) as surveyed in the introduction. A key innovation in our algorithm is that we incorporate consistency equations into the system of equations and unknowns so the transition of the endogenous state variable  $\omega_{t+1}^1$  is solved jointly with policy variables. These equations require that the conjectured future endogenous state variables are consistent with the current portfolio choices and future prices:

$$\omega_{t+1}^1 = \frac{(\hat{q}_{t+1}(z_{t+1}, \omega_{t+1}^1) + d_{t+1})k_{t+1}^1 + \hat{b}_{t+1}^1/g_{t+1}}{\hat{q}_{t+1}(z_{t+1}, \omega_{t+1}^1) + d_{t+1}}.$$

We discuss these equations in more detail in Subsection 3.2 in the context of our general framework. The GDSGE code for the model that implements our algorithm is given below (the consistency equations correspond to line 68 in the GDSGE code).

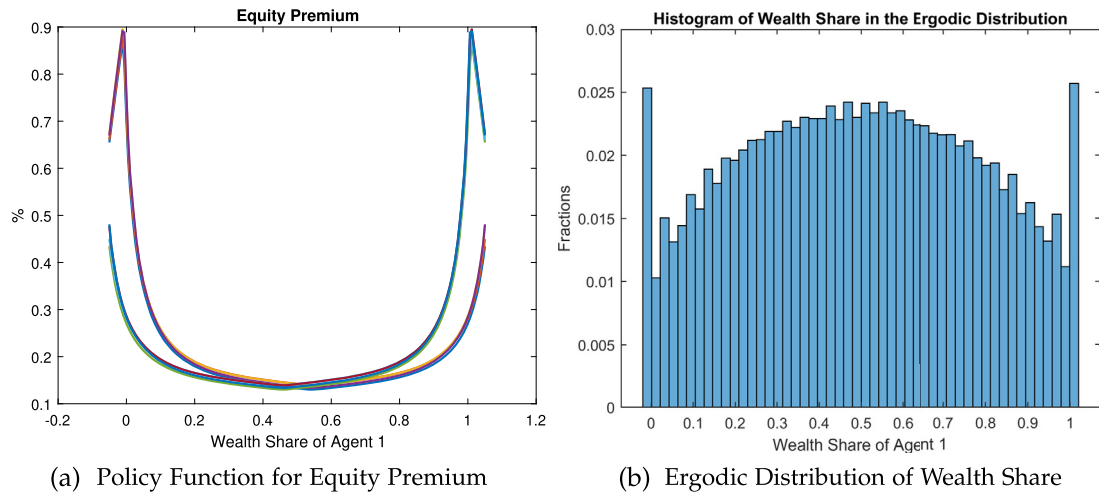
```

1 % Parameters
2 parameters beta gamma Kb;
3 beta = 0.95; % discount factor
4 gamma = 1.5; % CRRA coefficient
5 Kb = -0.05; % borrowing limit in ratio of aggregate output
6 % Exogenous state variables
7 var_shock g d etal;
8 % Enumerate exogenous states and transition matrix
9 shock_num = 8;
10 g = [.9904 1.0470 .9904 1.0470 .9904 1.0470 .9904 1.0470];
11 d = [.1402 .1437 .1561 .1599 .1402 .1437 .1561 .1599];
12 etal = [.3772 .3772 .3772 .3772 .6228 .6228 .6228 .6228];
13 shock_trans = [
14 0.3932 0.2245 0.0793 0.0453 0.1365 0.0779 0.0275 0.0158
15 0.3044 0.3470 0.0425 0.0484 0.1057 0.1205 0.0147 0.0168
16 0.0484 0.0425 0.3470 0.3044 0.0168 0.0147 0.1205 0.1057
17 0.0453 0.0793 0.2245 0.3932 0.0157 0.0275 0.0779 0.1366
18 0.1366 0.0779 0.0275 0.0157 0.3932 0.2245 0.0793 0.0453
19 0.1057 0.1205 0.0147 0.0168 0.3044 0.3470 0.0425 0.0484
20 0.0168 0.0147 0.1205 0.1057 0.0484 0.0425 0.3470 0.3044
21 0.0158 0.0275 0.0779 0.1365 0.0453 0.0793 0.2245 0.3932
22 ];
23 % Endogenous state variables
24 var_state w1; % wealth share
25 w1 = linspace(-0.05,1.05,201);
26 % Policy variables and bounds that enter the equations
27 var_policy c1 c2 slp nb1p nb2p ms1 ms2 mb1 mb2 ps pb wln[8];
28 inbound c1 0 1;
29 inbound c2 0 1;
30 inbound slp 0.0 1.0;
31 inbound nb1p 0.0 1.0; % nb1p=blp-Kb
32 inbound nb2p 0.0 1.0;
33 inbound ms1 0 1; % Multipliers for constraints
34 inbound ms2 0 1;
35 inbound mb1 0 1;
36 inbound mb2 0 1;
37 inbound ps 0 1 adaptive(1.5);
38 inbound pb 0 1 adaptive(1.5);
39 inbound wln -0.5 1.5;
40 % Other policy variables
41 var_aux equity_premium;
42 % Interpolation variables for state transitions
43 var_interp ps_future c1_future c2_future;
44 initial ps_future 0.0;
45 initial c1_future w1.*d+etal;
46 initial c2_future (1-w1).*d+1-etal;
47 ps_future = ps;
48 c1_future = c1;
49 c2_future = c2;
50
51 model;
52 % Evaluate interpolation
53 [psn',c1n',c2n'] = GDSGE_INTERP_VEC'(w1n');
54 % Calculate expectations that enter the Euler Equations
55 es1 = GDSGE_EXPECT{g'^(1-gamma)*(c1n'/c1)^(-gamma)*(psn'+d')/ps};
56 es2 = GDSGE_EXPECT{g'^(1-gamma)*(c2n'/c2)^(-gamma)*(psn'+d')/ps};
57 eb1 = GDSGE_EXPECT{g'^(1-gamma)*(c1n'/c1)^(-gamma)/pb};
58 eb2 = GDSGE_EXPECT{g'^(1-gamma)*(c2n'/c2)^(-gamma)/pb};
59 % Transform bond back
60 b1p = nb1p + Kb;
61 b2p = nb2p + Kb;
62 % Market clearing of shares
63 s2p = 1-s1p;
64 % Budget constraints
65 budget_1 = w1*(ps+d)+etal - c1 - ps*s1p - pb*b1p;
66 budget_2 = (1-w1)*(ps+d)+(1-etal) - c2 - ps*s2p - pb*b2p;
67 % Consistency equations
68 w1_consist' = (slp*(psn'+d') + b1p/g')/(psn'+d') - w1n';
69 % Other policy variables
70 equity_premium = GDSGE_EXPECT{(psn'+d')/ps*g'} - 1/pb;
71 equations;
72 -1+beta*es1+ms1;
73 -1+beta*es2+ms2;
74 -1+beta*eb1+mb1;
75 -1+beta*eb2+mb2;
76 ms1*s1p;
77 ms2*s2p;
78 mb1*nb1p;
79 mb2*nb2p;
80 b1p+b2p;
81 budget_1;
82 budget_2;
83 w1_consist';
84 end;
85 end;
86
87 simulate;
88 num_periods = 10000;
89 num_samples = 24;
90 initial w1 0.5;
91 initial shock 1;
92 var_simu c1 c2 ps pb equity_premium;
93 w1' = w1n';
94 end;

```

The code runs in less than a minute on a regular laptop (about 50 seconds on a MacBook Air with a 2.2 GHz Intel Core i7 CPU or 7 seconds on a desktop with Intel 2.5 GHz 12-Core CPU). It produces the policy functions, including equilibrium prices and allocations as functions of the endogenous state variable, agent 1's wealth share  $\omega^1$ , and exogenous state variable  $z$ . Panel (a) in Fig. 1 shows the equity premium (the difference between expected stock and bond returns) as a function of wealth share and for different combination of exogenous state variables. The kinks in the equity premium function appear at points where the borrowing or short-sale constraints switch from being binding to non-binding, or vice versa, as  $\omega^1$  increases. Panel (b) in Fig. 1 shows the ergodic distribution of the endogenous state variable,  $\omega^1$ .

The model can also be solved using consumption share instead of wealth share as a state variable, as in Dumas and Lyasoff (2012). In this case, the consistency equations correspond to agents' future budget constraints: future consumption shares should be consistent with current portfolio choices and future portfolio choices, which in turn depend on future consumption shares. Dumas and Lyasoff (2012) call these equations "marketability conditions." Our algorithm is more general and does not rely on their "kernel conditions" which are derived by assuming that the agents' unconstrained Euler equations hold exactly. Our algorithm allows for deviation from the Euler equations due to binding portfolio constraints, such



**Fig. 1.** Policy functions and ergodic distribution. Note: The model is solved with 8 realizations of the exogenous states and 201 grid points for the endogenous state. The histogram is based on 24 sample paths and 10,000 periods per sample path, with the first 1,000 periods dropped.

**Table 1**

Accuracy and speed of the algorithm and comparisons with existing methods.

|     | Specification                 | number of collocation points | Running time (seconds)     | Mean errors | Max errors |
|-----|-------------------------------|------------------------------|----------------------------|-------------|------------|
| (1) | STPFI                         | 1608                         | 7                          | 2.08E-05    | 3.40E-03   |
| (2) | STPFI, dense grid             | 8040                         | 24                         | 9.09E-07    | 5.91E-04   |
| (3) | STPFI, adaptive grid          | 1548                         | 11                         | 9.22E-06    | 5.78E-04   |
| (4) | Transition Function Iteration | 1608                         | 37                         | 2.61E-05    | 3.38E-03   |
| (5) | NFP-Guvenen                   | 1608                         | 108 (converged under 5e-5) | 2.25E-05    | 3.33E-03   |
| (6) | NFP-KS                        | 1608                         | 11                         | 2.08E-05    | 3.40E-03   |

Note: Running time is based on a desktop with an Intel 2.5 GHz 12-Core CPU. The mean and max Euler equation errors are calculated across 24 sample paths and 10,000 periods per sample path, with the same random number draws across methods. All algorithms are implemented carefully so the performances can be fairly compared (see the text for detail). All algorithms, except Algorithm (5), converge under the criterion that the maximum absolute distance of policy functions across allocation points is below the metric of  $1E-6$  between two adjacent iterations. Algorithm (5) can only converge under the metric of  $5e-5$ . STPFI corresponds to the *Simultaneous Transition and Policy Function Iteration* algorithm developed in the current paper. Transition Function Iteration (TFI) corresponds to the algorithm developed by Elenev et al. (2021), which at each time step fixes the transition function constructed from the previous iteration. NFP-Guvenen solves the recursive equilibrium using a nested loop: with the inner loop for solving policy functions until convergence and outer loop for solving the transition functions. NFP-KS solves the equilibrium system at each time step in a nested manner, with the inner nest for future endogenous states and outer nest for policy variables. Algorithms (1) and (4)–(6) use the same grid for collocation points and cubic splines; (2) uses a denser grid; (3) uses an adaptive grid following Brumm and Scheidegger (2017) implemented by the toolbox.

as borrowing constraints or short-sale constraints. The details of our GDSGE implementation are provided on the toolbox's website. In Section 4.1, we also show how to simplify the feasible region of the endogenous state-space in Mendoza (2010) using consumption as one endogenous state variable.

To assess the accuracy and speed of our algorithm and the toolbox compared with existing methods, we record the running time of solving the model with different specifications. We calculate the unit-free Euler equation errors for shares and bonds of agent  $i \in \{1, 2\}$  for a large sample drawn from the model's ergodic distribution<sup>15</sup>:

$$\begin{aligned}\mathcal{E}_t^{s,i} &= -1 + \beta \mathbb{E}_t \left[ \left( \frac{\hat{c}_{t+1}^i}{\hat{c}_t^i} \right)^{-\gamma} (\gamma_{t+1}^a)^{1-\gamma} \frac{\hat{p}_{t+1}^s + \hat{d}_{t+1}}{\hat{p}_t^s} \right] + \hat{\mu}_t^{i,s}, \\ \mathcal{E}_t^{b,i} &= -1 + \beta \mathbb{E}_t \left[ \left( \frac{\hat{c}_{t+1}^i}{\hat{c}_t^i} \right)^{-\gamma} (\gamma_{t+1}^a)^{-\gamma} \frac{1}{\hat{p}_t^b} \right] + \hat{\mu}_t^{i,b}.\end{aligned}$$

We then calculate the mean and max errors across all samples.

Table 1 reports the running time and mean and max Euler equation errors for different algorithms and compare them. Algorithm (1) corresponds to our benchmark STPFI algorithm which is implemented by the toolbox code listed before and produces the output in Fig. 1. The algorithm uses a grid over  $\omega_1$  with 201 points and uses cubic splines to approximate the

<sup>15</sup> To do so, we simulate 24 sample paths with 11,000 periods per sample path and drop the first 1000 periods of each sample path to collect a sample of size  $24 \times 10,000$ .

policy and transition functions. The mean and max unit-free Euler equation errors are around  $2.1\text{E-}05$  and  $3.4\text{E-}03$ , which transform to \$0.14 and \$23 in \$10,000 consumption levels, respectively.<sup>16</sup> By increasing the number of grid points by 5 times, the Euler equation errors can be reduced by an order of magnitude, as shown in algorithm (2). However, this can be done more efficiently by using an adaptive-grid approach developed by Ma and Zabarar (2009) and Brumm and Scheidegger (2017). Our toolbox implements the algorithm efficiently and, as shown in algorithm (3), reduces the (max) Euler equation errors by an order of magnitude, without increasing the number of grid points or increasing the running time significantly.<sup>17</sup>

We compare the performance of our algorithm and an alternative algorithm (4), TFI, used in Elenov et al. (2021). This algorithm is also based on time iterations, but unlike ours which solves the transition and policy functions jointly in each iteration, theirs fixes the state transition function implied by the policy functions solved from the previous iteration.<sup>18</sup> We implement this alternative algorithm with the same numerical routines (including the equation solver and function approximation procedure) in C++, starting from the same initial transition function,<sup>19</sup> and fine-tuned for maximum numerical efficiency<sup>20</sup> so the performance of the two algorithms can be compared fairly. Using dampened updating with a damping parameter of 0.25 (i.e., with weight 0.75 assigned to updated transition functions), algorithm (4) can converge successfully, and the Euler equation errors are comparable to STPFI. The speed is significantly slower than STPFI, mainly because their algorithm fixes the state transition function and does not allow the future endogenous state (in the current example, future wealth share) to respond to current policy variables (share and bond choices etc.), which makes finding a solution to the equilibrium system more difficult.<sup>21</sup>

The second alternative algorithm, (5) in the table, is the nested fixed point algorithm. The algorithm proceeds in a nested loop, similar to the algorithm in Krusell and Smith (1998) for fully heterogeneous agent models, with the inner loop solving for converged policy functions fixing the state transition function and the outer loop solving for the state transition function that is consistent with the one implied by the inner loop. We find that such an algorithm is very unstable, and can only start to converge with a sufficient large dampening parameter for updating the transition function, especially for early iterations. Even with a large dampening parameter (0.75) the algorithm only converges with the higher threshold of  $5\text{e-}5$  (measured by the metric between policy functions of adjacent iterations), although the Euler equation errors are close to those of the STPFI benchmark. Due to its nested nature, the speed is also significantly slower than STPFI. Guvenen (2009) uses a similar nested fixed point algorithm for solving an asset pricing model with a richer setup. We demonstrate how to solve the model in the toolbox and discuss the performance comparison in Section 4.3.

The third alternative algorithm, (6) in the table, is the one developed by Kubler and Schmedders (2003) and is labeled as NFP-KS. Like STPFI, their algorithm solves policy variables and future state variables at each time step that respect all short-run equilibrium conditions but does so in a nested manner: the inner loop solves for the future state variables to satisfy the consistency equations with the current policy variables fixed, and the outer loop solves for the current policy variables that satisfy the remaining equilibrium conditions. As shown in the table, the algorithm is slower than STPFI, but once converged under the same criterion, delivers solutions with similar accuracies. NFP-KS is slower because it incurs substantially higher number of evaluations of the consistency equations, which outweighs the benefits of solving a smaller system of equations each time. Lastly, this algorithm is also not amenable to an automated design. The performance in the reported table requires manually incorporating into the toolbox-generated C++ code an inner loop that solves the consistency equations using the Newton method. Implementation details such as initial guesses are critical for the performance. Therefore, although the performance does not deteriorate much compared to STPFI, the algorithm requires substantially more manual efforts and trial and error, as with any other nested algorithms. A simple and robust implementation demonstrates the appeal of using a non-nested algorithm such as STPFI.

On the toolbox website, we provide a further breakdown of the computation costs of different algorithms.<sup>22</sup> The breakdown shows that the STPFI algorithm is more efficient for the current model since it balances the number of evaluations of intra-temporal conditions (i.e., equilibrium conditions that only involve current-period variables such as market-clearing conditions and budget constraints), inter-temporal conditions (i.e., Euler equations), and consistency equations. Other algorithms instead are designed to reduce the number of evaluations of inter-temporal conditions, or reduce the system

<sup>16</sup> The unit-free errors can be transformed to errors in consumption levels by multiplying them by  $1/\gamma$ .

<sup>17</sup> See Section 4.4 for another example which benefits even more from the adaptive-grid method due to the strong state-dependence of the model.

<sup>18</sup> Their algorithm can be also implemented using our toolbox. A detailed description of the algorithm and the implementation with the toolbox can be found in Online Appendix A.1, in particular footnote 32. We provide a more detailed comparison in the context of this Heaton and Lucas's model on the toolbox website: <https://www.gdsge.com/example/HL1996/HL1996TFIter.html>.

<sup>19</sup> The initial transition function is generated by the solution to the last-period problem of a finite-horizon economy.

<sup>20</sup> For example, for the alternative algorithms we pre-compute the expectation terms when they do not depend on current policy variables under the fixed state transition functions. For the nested fixed point algorithms we use the solution of policy functions of the previous transition function iteration as “warm-ups” to increase the speed of convergence in the inner loop.

<sup>21</sup> Another reason why their algorithm is slower is because their algorithm needs to first evaluate the state transition function to get future states and then evaluate the policy functions to get future allocations and prices, whereas STPFI only needs to evaluate the policy functions as future endogenous states are solved simultaneously. However, the speed difference due to this reason is mitigated by that the future states and expectation terms can be pre-computed before solving the equilibrium system as they do not vary with current policy variables.

<sup>22</sup> The detailed breakdown and discussions can be found at <http://www.gdsge.com/example/HL1996/HL1996.html#comparison-with-alternative-algorithms>.



of equations solved simultaneously, and may demonstrate their strength more visibly in models where evaluating intertemporal conditions is more costly.

In the next section we present the general framework beyond the specific example from Heaton and Lucas (1996).

### 3. General environment

In this section, we provide the general framework and the solution algorithm (STPFI) to compute recursive equilibrium in this framework. In Online Appendix A, we present the design of the toolbox to implement the algorithm. In Section 4, we show that many recent important models fit exactly in the framework and hence can be solved using the toolbox, with an underlying algorithm that is different from the algorithms in the original papers.

#### 3.1. Recursive equilibrium and solution algorithm

We work with models for which the sequential competitive equilibrium of the economy can be characterized by a system of short-run equilibrium conditions:

$$\mathbf{F}(s, x, z, \{s'(z'), x'(z')\}_{z' \in \mathcal{Z}}) = 0 \quad (1)$$

where  $z \in \mathcal{Z} \subset \mathbb{R}^{d_z}$  is a vector of exogenous shocks (aggregate productivity, aggregate growth rates, dividends, etc.),  $s \in \mathcal{S} \subset \mathbb{R}^{d_s}$  is a vector of endogenous state variables (aggregate capital, wealth shares, consumption shares, etc.), and  $x \in \mathcal{X} \subset \mathbb{R}^{d_x}$  is a vector of endogenous policy variables (asset prices, consumption, portfolio choices, Lagrange multipliers, etc.). The function

$$\mathbf{F} : \mathbb{R}^{d_s+d_x+d_z} \times \left( \mathbb{R}^{d_s} \times \mathbb{R}^{d_x} \right)^Z \Rightarrow \mathbb{R}^{d_s+d_x+d_z} \times \left( \mathbb{R}^{d_s} \times \mathbb{R}^{d_x} \right)^Z,$$

where  $Z$  is the cardinality of  $\mathcal{Z}$ , consists of optimality conditions, market clearing conditions, and laws of motion for state variables. For example, in the model in Heaton and Lucas (1996) described above  $z = (\gamma^a, \delta, \eta)$ ,  $s = (\omega^1)$ , and  $x = (\hat{c}^1, s^1, \hat{b}^1, \hat{c}^2, s^2, \hat{b}^2, p^s, p^b)$ .

Notice that the framework allows for general dependence on the future variables, instead of through common expectations as in Winschel and Kratzig (2010). This generality is important to include models with state variables with implicit law of motion. We discuss this feature in more detail in Subsection 3.2. It also allows for non-rational expectations models including model with belief heterogeneity such as Sandroni (2000) and Cao (2018). Lastly, it is necessary to capture non-linear forms of borrowing constraints including the collateral constraints in Kiyotaki and Moore (1997) and Cao and Nie (2017).<sup>23</sup>

Models with inequality constraints fit into the general formulation (1) by adding additional endogenous policy functions. Indeed, if a recursive model has both equality and inequality conditions (such as the borrowing constraints in Heaton and Lucas (1996)),

$$\begin{aligned} \mathbf{F}(s, x, z, \{s'(z'), x'(z')\}_{z' \in \mathcal{Z}}) &= 0 \\ \mathbf{G}(s, x, z, \{s'(z'), x'(z')\}_{z' \in \mathcal{Z}}) &\geq 0, \end{aligned}$$

we can use  $\hat{\mathbf{F}} = \begin{pmatrix} \mathbf{F} \\ \mathbf{G} - \eta \end{pmatrix}$  with  $\eta \geq 0$ ,<sup>24</sup> and  $\hat{x} = (x, \eta)$ , to write the system with inequality constraint in form (1) for  $\hat{\mathbf{F}}$  and  $\hat{x}$ .

In this general framework, we look for recursive equilibria defined below.

**Definition** A recursive equilibrium is a solution to (1) of the form

$$x = \mathcal{P}(z, s)$$

and

$$s'(z') = \mathcal{T}(z, z', s)$$

where  $\mathcal{P}$  and  $\mathcal{T}$  are equilibrium policy and transition functions, respectively.

<sup>23</sup> Collateral constraints might involve nonlinear functions of future asset prices (as random variables) beyond simple functions of expected prices, such as the minimum of the price realizations over all possible future states. Cao and Nie (2017) provide a detailed comparison of different forms of collateral constraints.

<sup>24</sup> In numerical implementations, we tackle this with an equation solver that respects box constraints of unknowns.

*The Simultaneous Transition and Policy Function Iteration (STPFI) algorithm* We solve for a recursive equilibrium of (1) using STPFIs as follows. The algorithm starts with an initial guess for policy and transition functions

$$\left\{ \mathcal{P}^{(0)}(., .), \mathcal{T}^{(0)}(., ., .) \right\}.$$

Given  $\mathcal{P}^{(n)}$  and  $\mathcal{T}^{(n)}$ ,  $\mathcal{P}^{(n+1)}$  and  $\mathcal{T}^{(n+1)}$  are determined by solving the following system of equations:

$$\mathbf{F}\left(s, x, z, \left\{s'(z'), \mathcal{P}^{(n)}(z', s'(z'))\right\}_{z' \in \mathcal{Z}}\right) = 0, \quad (2)$$

with unknowns  $x$  and  $\{s'(z')\}_{z' \in \mathcal{Z}}$  for each

$$(z, s) \in \mathcal{C}^{(n)} \subset \mathcal{Z} \times \mathcal{S}.$$

The set  $\mathcal{C}^{(n)}$ , which we call the set of collocation points, is a subset of  $\mathcal{Z} \times \mathcal{S}$ . We keep track of a distance between  $\mathcal{P}^{(n)}$ ,  $\mathcal{T}^{(n)}$  and  $\mathcal{P}^{(n+1)}$ ,  $\mathcal{T}^{(n+1)}$  over the iterations and stop when the distance falls below a preset threshold.

Unlike value function iterations for Bellman equations for which the Contraction Mapping Theorem applies, good initial guesses are critical in ensuring that the algorithm works. Otherwise one might run into situations in which (2) does not have solution for some  $(s, z) \in \mathcal{C}^{(n)}$  and hence cannot iterate further, or the iterations do not converge. We find that good typical initial guesses for  $\mathcal{P}^{(0)}$  correspond to equilibria in the 1-period economy versions of the models. In this case, the solution for  $\mathcal{P}^{(n)}$  corresponds to the equilibrium values of the first period in the  $(n+1)$ -period economy. Thus, the numerical limit of  $\{\mathcal{P}^{(n)}\}$  corresponds to the finite-horizon limit. This limit is shown to be the equilibrium in the infinite horizon economies in existence proofs for infinite-horizon incomplete markets economies such as Duffie et al. (1994), Magill and Quinzii (1994), and more recently Cao (2020).

### 3.2. Explicit and implicit state transitions

Our toolbox demonstrates its greatest power, relative to other methods, for models with endogenous state variables with implicit state transition equations (laws of motion), such as wealth shares or consumption shares in the leading Heaton and Lucas' example. Here we explain the idea in more detail in the context of the general framework.

The state variables  $s$  may consist of state variables  $\bar{s}$  that have explicit transition equations (laws of motions), and state variables  $\tilde{s}$  with implicit transition equations:  $s = (\bar{s}, \tilde{s})$ . For  $\tilde{s}$ , the law of motion can be written explicitly:

$$\tilde{s}' = \bar{g}(s, x, z, z'),$$

which is the specification in Winschel and Kratzig (2010). Our framework allows for state variables  $\tilde{s}$  with implicit laws of motion:

$$0 = \bar{g}(s, x, z, \tilde{s}'(z'), x'(z'), z').$$

Examples of state variables with implicit state transition include wealth shares or consumption shares, as in Section 2 for Heaton and Lucas (1996).

In this case, the system of equations (1) can be written as

$$\mathbf{F}(s, x, z, \{s'(z'), x'(z')\}_{z' \in \mathcal{Z}}) = \begin{pmatrix} f(s, x, z, \{s'(z'), x'(z')\}_{z' \in \mathcal{Z}}) \\ \tilde{s}' - \bar{g}(s, x, z, z') \\ \bar{g}(s, x, z, \tilde{s}'(z'), x'(z'), z') \end{pmatrix}.$$

In a recursive equilibrium, the last equation becomes

$$0 = \bar{g}(s, x, z, \tilde{s}'(z'), \mathcal{P}(z', (\bar{g}(s, x, z, z'), \tilde{s}'(z'))), z'). \quad (3)$$

We call these equations *consistency equations*. It requires future state variables  $\tilde{s}'(z')$  to be consistent with current policies and future policies implied by these future state variables and the policy function  $\mathcal{P}$ .<sup>25</sup>

The state variables with explicit state transitions allow us to reduce the number of equations and unknowns in each step of the policy function iteration algorithm described above. Indeed, by substituting  $\bar{g}(s, x, z, z')$  for  $\tilde{s}'$ , we can work with  $\bar{\mathbf{F}}$  which only takes the first and third components from  $\mathbf{F}$ :

<sup>25</sup> These consistency equations are similar to the consistency equations in the heterogeneous agent incomplete markets literature, including Huggett (1993), Huggett (1997), Aiyagari (1994), Krusell and Smith (1998), Reiter (2009) and Cao (2020), in which future cross-sectional distributions are consistent with current distributions and current policy functions. Reiter (2015) explicitly imposes consistency equations in an OLG model with cohorts' net worths as endogenous state variables but he does not include the net worths among the unknowns to be solved in each iteration like we do. This seemingly minor detail greatly facilitates the automation of our algorithm.

$$\bar{F}\left(s, x, z, \left\{\bar{s}'(z'), \mathcal{P}^{(n)}\left(z', (\bar{g}(s, x, z, z'), \bar{s}'(z'))\right)\right\}_{z' \in \mathcal{Z}}\right) = 0.$$

The state variables with implicit laws of motion make it difficult to apply standard policy function iteration methods. Because one cannot easily calculate future state variables, and hence future equilibrium variables, to evaluate inter-temporal equilibrium conditions. To tackle this issue, the key innovation in our algorithm is, in  $(n+1)^{\text{th}}$ -iteration, to simultaneously solve for the unknowns  $x$  and  $\{\bar{s}'(z')\}_{z' \in \mathcal{Z}}$  given future policy function  $\mathcal{P}^{(n)}$ . To do so, we add to the system of equations (2) the consistency equations (3):

$$\bar{g}\left(s, x, z, \bar{s}'(z), \mathcal{P}^{(n)}\left(z', (\bar{g}(s, x, z, z'), \bar{s}'(z'))\right), z'\right) = 0.$$

As discussed in the introduction, our approach to solving models with endogenous state variables with implicit laws of motion is different from the existing approaches in the literature. For example, Kubler and Schmedders (2003) use wealth shares as endogenous state variables. These authors solve for future wealth shares using consistency equations as an additional fixed-point problem for each guess for current policy variables. The solution to the fixed-point problem is then used to formulate a system of equations and unknowns for current policy variables. This nested-fixed point procedure might be unstable. For example, the nested fixed point problems might not have a solution in early iterations when the initial guess for policy functions is far from the converged functions. In addition, the procedure is not amenable to a simple, automated toolbox implementation because the systems of equations representing the nested-fixed point problems are completely different from the original equilibrium systems of equations. Elenev et al. (2021)'s method iterates over transition functions. As described in Section 2, this method can be implemented in our toolbox but it might not be as accurate and robust as our method. One of the possible reasons for lower accuracy is that the approach does not allow for feedback from future variables on current policy variables within each iteration, even though some of the feedback effect can be introduced by manually setting dampening parameters.

One potential concern with our method is that if the number of possible realizations of future exogenous shocks  $z'$  is too large, including  $\{\bar{s}'(z')\}_{z' \in \mathcal{Z}}$  and consistency equations in the system of equations and unknowns to be solved leads to a system that is too large. For example, if the true exogenous shocks  $\zeta$  follow a VAR process

$$\zeta' = A\zeta + \epsilon',$$

one needs to approximate this process with a discrete-Markov process  $z$  with many points. To deal with this issue, we include  $\zeta$  among the *endogenous* state variables  $s$  and discretize the innovation process  $\epsilon'$  instead, which requires a smaller number of discretization points and hence a smaller number of consistency equations.<sup>26</sup>

In Online Appendix A, we present the design and implementation of our toolbox for the general framework described above. In the next section, we present several important applications. Besides Heaton and Lucas (1996), the models from Mendoza (2010) and Barro et al. (2017) in Subsections 4.1 and 4.2 are other examples featuring state variables with implicit laws of motion.

#### 4. Applications

In this section, we provide examples of how well-known models can be solved using our toolbox. The gmod files for these models are provided in Online Appendix B and on the toolbox's website. The toolbox algorithm is different from the algorithm provided in the original papers. These examples could be read independently and the notation follows closely from the notation in the original papers. We also refer readers to the original papers for the important economic motivation of these models. Following up on the leading example from Heaton and Lucas (1996), the next two examples from Mendoza (2010) and Barro et al. (2017) feature state variables with implicit laws of motion and consistency equations.

##### 4.1. Sudden stops with asset price deflation by Mendoza (2010)

The seminal work by Mendoza (2010) builds a model that generates crises featuring current account reversals and asset price collapses, resembling those experienced by many emerging economies. The model has two endogenous assets—capital and bond, with an occasionally binding borrowing constraint tied to the capital price behind its key mechanism. The model is highly non-linear with state-dependent responses of endogenous variables to exogenous shocks, and calls for a global solution method to study its rich dynamics. Although it has become a workhorse model in the study of crises in open and closed economies, researchers still find solving and extending this type of model challenging. We show that the model in Mendoza (2010) can be easily represented in our current framework, and solved with the toolbox in 200 lines of codes and within a minute on a regular laptop (about 55 seconds on a MacBook Air 2.2 GHz Intel Core i7).

Even though with the benchmark parameterization in Mendoza (2010), the model can be solved with capital and bond as the two endogenous states directly, one of the challenges researchers have frequently countered is that due to the borrowing

<sup>26</sup> See the RBC model with irreversible investment on the toolbox's website (<http://www.gdsge.com/example/rbc/rbclrr.html>) for a concrete example.

constraint tied to the capital price, the feasible state space of capital and bond is not a rectangle and hard to determine ex-ante.<sup>27</sup> Our framework and toolbox address this challenge by transforming the endogenous states to those that have simple boundary conditions, and use *consistency equations* to specify the implicit transitions of these transformed states. This strategy frees researchers from trial-and-error guesses for the feasible space of states, a procedure that is necessary but usually painful for global solution methods. We first briefly describe the model and then its representation in our framework. As for other examples, the toolbox code is provided in Online Appendix B and can be downloaded from the GDSGE website.

#### 4.1.1. Model description

Representative consumers in a small-open economy value consumption and leisure, with preferences represented by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u(c_t - N(L_t))],$$

where  $\beta \in (0, 1)$  is the discount factor,  $u(\cdot)$  is the GHH (Greenwood et al., 1988) utility function,  $c_t$  is consumption,  $L_t$  is the labor supply, and  $N(\cdot)$  is a convex function representing the disutility from labor supply.  $\mathbb{E}_0$  is the expectation operator integrating aggregate shocks introduced below.

Output is produced by representative firms combining capital  $k_t$ , labor  $L_t$ , and imported inputs  $v_t$  according to a constant-return-to-scale production function

$$\exp(\varepsilon_t^A) F(k_t, L_t, v_t),$$

where  $\varepsilon_t^A$  is the aggregate productivity shock. Investment is subject to convex adjustment costs

$$i_t = k_{t+1} - (1 - \delta)k_t + \frac{1}{2}a \frac{(k_{t+1} - k_t)^2}{k_t}, \quad (4)$$

where  $a > 0$  is a parameter determining the level of adjustment cost. The end-of-period capital price is competitively determined as

$$q_t = \frac{\partial i_t}{\partial k_{t+1}} = 1 + a \frac{k_{t+1} - k_t}{k_t}. \quad (5)$$

A fraction of the cost of labor and imported inputs needs to be financed by within-period loans obtained from foreign lenders, at the world gross interest rate of  $R_t = R \exp(\varepsilon_t^R)$ , where  $R$  is the mean interest rate and  $\varepsilon_t^R$  is an exogenous shock. The price of imported goods is  $p_t = p \exp(\varepsilon_t^P)$ , where  $p$  is the mean price and  $\varepsilon_t$  is an exogenous shock.  $(\varepsilon_t^A, \varepsilon_t^R, \varepsilon_t^P)$  are the shocks in the economy and follow first-order Markov processes.

Consumers can trade a one-period state non-contingent bond  $b_{t+1}$  (a negative value of  $b_{t+1}$  denotes borrowing) with the rest of the world, at price  $q_t^b = \frac{1}{R_t}$ . Therefore, the budget constraint of the consumers reads:

$$c_t + i_t = \exp(\varepsilon_t^A) F(k_t, L_t, v_t) - p_t v_t - \phi(R_t - 1)(w_t L_t + p_t v_t) - q_t^b b_{t+1} + b_t,$$

of which the left hand side is domestic consumption and investment, and the terms on the right hand side are in order (1) domestic gross output, (2) minus costs of imported goods, (3) minus interest payments from working capital loans, (4) minus costs of bond holding into the future period, and (5) plus current bond payments. It is assumed that the intertemporal borrowing is subject to a collateral constraint:

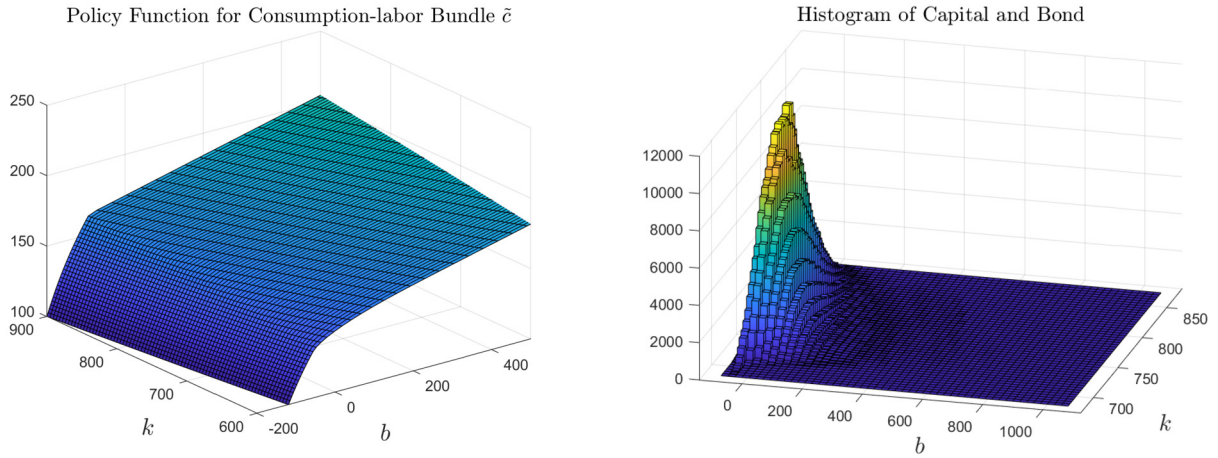
$$q_t^b b_{t+1} - \phi R_t (w_t L_t + p_t v_t) \geq -\kappa q_t k_{t+1},$$

which says that total debt, including debt in one-period bonds and working capital loans, does not exceed  $\kappa$  fraction of the market value of end-of-period capital. The representative consumers maximize the utility specified earlier, subject to the budget and collateral constraints described above. We next represent the model equilibrium with a system of equations, which are enabled by that the optimality conditions are replaced with first-order and complementary slackness conditions.

A sequential competitive equilibrium is stochastic sequences  $\{\mu_t, c_t, w_t, L_t, v_t, b_{t+1}, k_{t+1}\}_{t=0}^{\infty}$  such that

$$\begin{aligned} c_t + i_t &= \exp(\varepsilon_t^A) F(k_t, L_t, v_t) - p_t v_t - \phi(R_t - 1)(w_t L_t + p_t v_t) - q_t^b b_{t+1} + b_t, \\ -q_t^b \lambda_t + \mu_t q_t^b + \beta \mathbb{E}_t \lambda_{t+1} &= 0, \\ \beta \mathbb{E}_t \lambda_{t+1} \exp(\varepsilon_{t+1}^A) F_1(k_{t+1}, L_{t+1}, v_{t+1}) - \beta \mathbb{E}_t \lambda_{t+1} \frac{\partial i_{t+1}}{\partial k_{t+1}} - \lambda_t q_t + \mu_t \kappa q_t &= 0, \end{aligned}$$

<sup>27</sup> In particular, the maximum sustainable debt level that ensures positive consumption is increasing in capital stock, rendering the feasible space of capital and bond non-rectangular. Furthermore, this boundary cannot be determined ex-ante since the sustainable debt level depends on the new debt that the economy can finance, which depends on capital price and other *endogenous* variables determined in equilibrium.



**Fig. 2.** State transformation and the ergodic distribution. Note: The figure is generated using the baseline parameters in Mendoza (2010). The model is solved with 80 grid points over the endogenous state  $k$  and 80 grid points over the endogenous state  $\tilde{c}$ . The policy function corresponds to the first of the 8 realizations of exogenous shocks, i.e., all of  $\varepsilon^A$ ,  $\varepsilon^R$ ,  $\varepsilon^P$  take their lower values of the two realizations, respectively. The ergodic set is based on a simulated panel of 24 time series, each with 50,000 periods.

$$\begin{aligned}
 [q_t^b b_{t+1} - \phi R_t (w_t L_t + p_t v_t) + \kappa q_t k_{t+1}] \mu_t &= 0, \mu_t \geq 0, \\
 \exp(\varepsilon_t^A) F_2(k_t, L_t, v_t) &= w_t (1 + \phi(R_t - 1) + \frac{\mu_t}{\lambda_t} \phi R_t), \\
 \exp(\varepsilon_t^A) F_3(k_t, L_t, v_t) &= p_t (1 + \phi(R_t - 1) + \frac{\mu_t}{\lambda_t} \phi R_t), \\
 w_t &= N'(L_t),
 \end{aligned}$$

where,  $\lambda_t = u'(c_t - N(L_t))$ ,  $i_t$  is given by equation (4), and  $q_t$  is given by equation (5). Among the unknowns,  $\mu_t$  is the Lagrangian multiplier on the collateral constraint. In the system of equations, the first is the budget constraint; the second the Euler equation for bond; the third the Euler equation for capital; the fourth the complementary-slackness condition for the collateral constraint; the fifth and sixth are the optimality conditions for using labor and imported inputs, taking into account the interest payments on working capital loans and the shadow prices arising from the collateral constraint; the last is the optimality condition for labor supply. The equilibrium system can be cast in a recursive form with capital and bond as the endogenous states. Before describing the representation of the model using our framework, we discuss the state transformation and the associated consistency equations.

The motivation for state transformation, as described earlier, is that the feasible set of capital  $k_t$  and bond  $b_t$  may not be a rectangle, and the fact that the borrowing constraint depends on equilibrium capital price renders determining the feasible set ex-ante difficult. Instead, by noticing that the exact boundary condition of the feasible set is the consumption-labor bundle,  $\tilde{c}_t \equiv c_t - N(L_t)$ , being non-negative, we can transform the endogenous states from  $(k_t, b_t)$  to  $(k_t, \tilde{c}_t)$ . Unlike  $b_t$ , for which the period- $(t+1)$  bond is a policy variable and thus admits an explicit transition, the transition of  $\tilde{c}_t$  is implicitly imposed by the condition that  $\tilde{c}_{t+1}$  should be consistent with the choice of  $b_{t+1}$  via the future budget constraint. These conditions are precisely the *consistency equations* (equations (3) in the general framework) that should be included as part of the system of equations to be solved in each iteration. Therefore, with such transformation, the equation system described before should be modified to include the vector of  $\tilde{c}_{t+1}$  for each realization of future shocks as unknowns, and correspondingly to replace the current budget constraint with the future budget constraints for each realization of shocks.

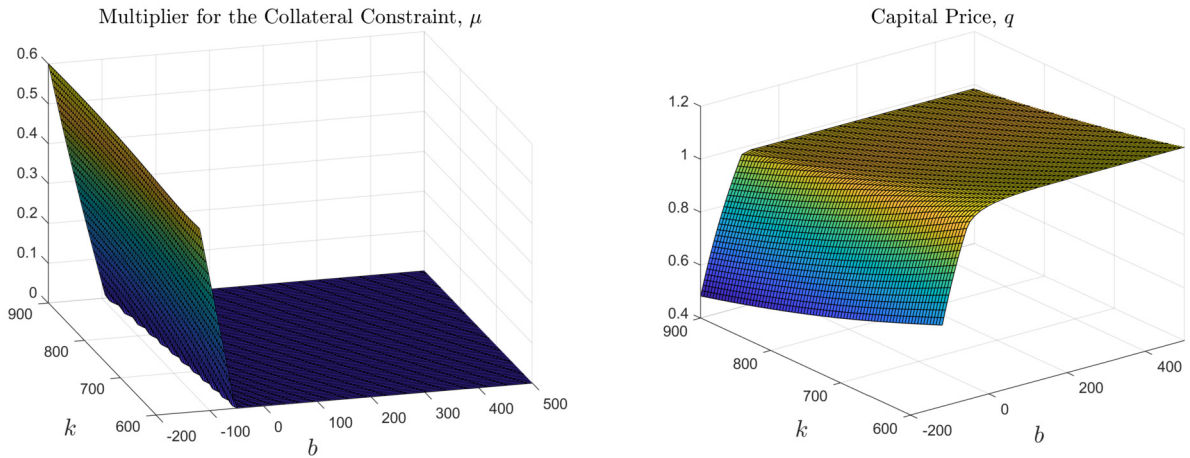
#### 4.1.2. Computation

We use the functional forms and parameterization from the benchmark calibration in Mendoza (2010).<sup>28</sup> The first set of results concerns about the motivation for the state transformation discussed earlier. The left panel of Fig. 2 projects the endogenous state  $\tilde{c}$ , over the natural state space  $(k, b)$ .<sup>29</sup> As shown in the left panel, for the rectangular state space defined over  $(k, \tilde{c})$ , the implied space of  $(k, b)$  is not a rectangle—in fact, the value of  $b$  that is consistent with the same value of  $\tilde{c}$  is increasing in  $k$ . Choosing a uniform lower bound for  $b$  too low may result in equilibrium non-existence for low values of  $k$ , since a positive consumption may not exist at such collocation points. On the other hand, choosing a uniform lower

<sup>28</sup> More detailed comparisons between the current results and those in Mendoza (2010), such as business cycle moments, can be found on the toolbox website.

<sup>29</sup> Notice that the equilibrium is solved over the endogenous states  $(k, \tilde{c})$ . For each combination of  $(k, \tilde{c})$  and realization of exogenous shocks, there is a value of  $b$  that is consistent with equilibrium conditions. We can therefore project policy functions defined over  $(k, \tilde{c})$  to the space of  $(k, b)$  as long as  $b$  is monotone in  $\tilde{c}$  given  $k$  and realizations of exogenous shocks, which is validated by the solution.





**Fig. 3.** Policy functions. Note: The figure is generated using the baseline parameters in Mendoza (2010). The model is solved with 80 grid points over the endogenous state  $k$  and 80 grid points over the endogenous state  $\tilde{c}$ . The policy functions are then generated by projecting the variables over the natural state space  $(k, b)$ . The policy functions correspond to the first of the 8 realizations of exogenous shocks, i.e., all of  $\varepsilon^A, \varepsilon^R, \varepsilon^P$  take their lower values of the two realizations, respectively.

bound for  $b$  too high may miss some combinations of  $(k, b)$  that appear in the model's ergodic set, as suggested by the right panel of Fig. 2, which shows that the economy spends significant fractions of time in the region with high levels of debt. The above discussions echo the challenge faced by researchers in choosing the state space of bonds without knowing the actual equilibrium properties, and highlight the usefulness of the state transformation strategy which is enabled by the consistency equation method in the current framework.

Fig. 3 plots the policy functions for capital price and the multiplier associated with the collateral constraint. As shown, the collateral constraint tends to bind when the debt level is high (bond is more negative) and capital is low; capital price drops sharply in debt level when the collateral constraint binds, while it drops only modestly when the collateral constraint does not bind. These nonlinear regions appear in the model's ergodic set as shown in Fig. 2, highlighting the importance of solving the current model with global methods.

#### 4.1.3. Mapping into the general setup

For the model in Mendoza (2010) described above, the correspondence with our general setup of the toolbox is  $z = (\varepsilon^A, \varepsilon^R, \varepsilon^P)$ ,  $s = (k, \tilde{c})$ , and  $x = (\mu, w, L, v, b', k')$ .

#### 4.2. Safe assets by Barro et al. (2017)

Barro et al. (2017) incorporate heterogeneous risk-aversion into the model with rare disasters in Barro (2006) to study the endogenous creation of safe asset. Their model features incomplete markets: agents can only trade a stock and a bond as in Heaton and Lucas (1996). They solve their model using a mixture of projection and perturbation methods developed in Fernández-Villaverde and Levintal (2018). Our toolbox's algorithm is a purely projection method. It uses wealth shares as endogenous state variables. As Barro et al. (2017) discuss in their paper, their solution method is not sufficiently accurate for large values of risk-aversion coefficients.<sup>30</sup> We show below that our method can tackle these cases effectively and uncover new economic insights in these cases. For the more risk-averse agent's risk-aversion coefficient of 50, our toolbox solves the model within a minute on a regular laptop (about 40 seconds on a MacBook Air 2.2 GHz Intel Core i7).

##### 4.2.1. Model and normalization

There are two groups of agents,  $i = 1, 2$  in the economy. Agents have an Epstein and Zin (1989)-Weil (1990) utility function. The coefficients of risk aversion satisfy  $\gamma_2 \geq \gamma_1 > 0$ , i.e., agents 1 are less risk-averse than agents 2. The other parameters between these two groups are the same. There is a replacement rate  $v$  at which each type of agents moves to a state that has a chance of  $\mu_i$  of switching into type  $i$ . Taking the potential type shifting into consideration, their utility function can be written as

$$U_{i,t} = \left\{ \frac{\rho + v}{1 + \rho} C_{i,t}^{1-\theta} + \frac{1-v}{1+\rho} \left[ \mathbb{E}_t \left( U_{i,t+1}^{1-\gamma_i} \right) \right]^{\frac{1-\theta}{1-\gamma_i}} \right\}^{1/(1-\theta)}. \quad (6)$$

<sup>30</sup> See Table 2 in their paper.

In this economy, there is a Lucas tree generating consumption good  $Y_t$  in period  $t$  consumed by both agents.  $Y_t$  is subject to identically and independently distributed rare-disaster shocks. With probability  $1 - p$ ,  $Y_t$  grows by the factor  $1 + g$ ; with a small probability  $p$ ,  $Y_t$  grows by the factor  $(1 + g)(1 - b)$ . Thus the expected growth rate of  $Y_t$  in each period is  $g^* \approx g - pb$ . Denote agent  $i$ 's holding of the tree as  $K_{it}$ . The supply of the Lucas tree is normalized to one;  $P_t$  denotes its price. The gross return of holding equity is  $R_t^e = \frac{Y_t + P_t}{P_{t-1}}$ . Agents also trade a risk-free bond,  $B_{it}$ , whose net supply is zero, and the gross interest rate is  $R_t^f$ .

Denote the beginning-of-period wealth of agent  $i$  by  $A_{it}$ . Each agent's budget constraint is

$$C_{it} + P_t K_{it} + B_{it} = A_{it}.$$

Considering the type shifting shock, the law of motion of  $A_{it}$  is

$$A_{it} = (Y_t + P_t) [K_{it-1} - \nu (K_{it-1} - \mu_i)] + (1 - \nu) R_t^f B_{it-1}.$$

As in Cao (2018, Appendix C.3, Extension 3), we normalize the utility  $U_{it}$  and consumption  $C_{it}$  by  $A_{it}$  and write equation (6) as follows:

$$u_{it}^{1-\theta} = \frac{\rho + \nu}{1 + \rho} c_{i,t}^{1-\theta} + \frac{1 - \nu}{1 + \rho} (1 - c_{it})^{1-\theta} \left( \mathbb{E}_t \left[ (R_{i,t+1} u_{it+1})^{1-\gamma_i} \right] \right)^{\frac{1-\theta}{1-\gamma_i}}, \quad (7)$$

in which  $u_{it} = U_{it}/A_{it}$ ,  $c_{it} = C_{it}/A_{it}$ , and

$$R_{i,t+1} = x_{it} R_{t+1}^e + (1 - x_{it}) R_{t+1}^f$$

is the average return of agent  $i$ 's portfolio, and

$$x_{it} = \frac{P_t K_{it}}{P_t K_{it} + B_{it}}$$

is the equity share of agent  $i$ 's portfolio holding. The FOCs for consumption and portfolio choices are

$$(\rho + \nu) c_{i,t}^{-\theta} = (1 - \nu) (1 - c_{it})^{-\theta} \left[ \mathbb{E}_t (R_{i,t+1} u_{it+1})^{1-\gamma_i} \right]^{\frac{1-\theta}{1-\gamma_i}}, \quad (8)$$

and

$$\mathbb{E}_t \left[ \frac{(R_{t+1}^e - R_{t+1}^f) u_{it+1}}{(R_{i,t+1} u_{it+1})^{\gamma_i}} \right] = 0. \quad (9)$$

The choices of  $c_{it}$  and  $x_{it}$  are identical across agents of the same type  $i$ , and the portfolio choice of agents  $i$  is

$$\begin{aligned} K_{it} &= x_{it} (1 - c_{it}) (1 + p_t) / p_t \omega_{it}, \\ B_{it} &= (1 - x_{it}) (1 - c_{it}) (1 + p_t) \omega_{it}. \end{aligned}$$

In equilibrium, prices are determined such that markets clear:

$$C_{1t} + C_{2t} = Y_t, \quad (10)$$

$$K_{1t} + K_{2t} = 1, \quad (11)$$

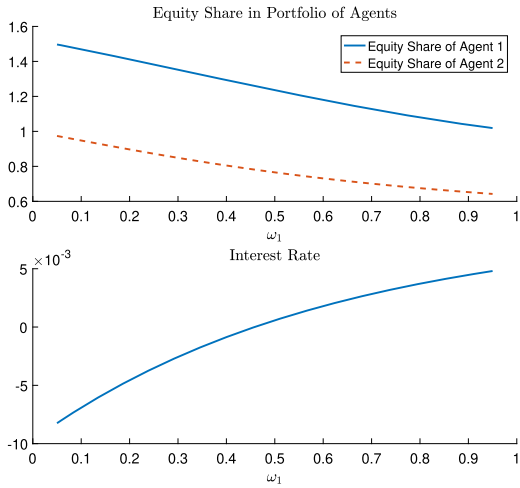
$$B_{1t} + B_{2t} = 0. \quad (12)$$

To achieve stationarity, we normalize variables  $\{B_{it}, P_t\}$  by  $Y_t$ . We define the wealth share of agent  $i$  as

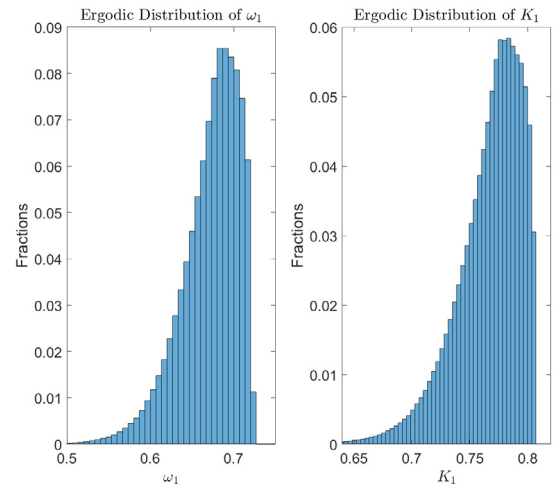
$$\omega_{it} = K_{it-1} - \nu (K_{it-1} - \mu_i) + \frac{(1 - \nu) R_t^f B_{it-1}}{(1 + p_t)(1 + g_t)}. \quad (13)$$

We see that given the market clearing conditions (11) and (12),

$$\omega_{1t} + \omega_{2t} = 1, \quad \forall t.$$



(a) Policy Functions



(b) Ergodic Distributions

**Fig. 4.** Ergodic distribution and policy functions. Note: The figure is generated using the baseline parameters in Barro et al. (2017). For annual data,  $\rho = 0.02$ ,  $\nu = 0.02$ ,  $\mu = 0.5$ ,  $\gamma_1 = 3.3$ , and  $\gamma_2 = 5.6$ . The growth rate in normal times is 0.025. Rare disasters happen with probability 4%, and once a rare disaster happens, productivity drops by 32%. The model period is one quarter.

#### 4.2.2. Log utility

For much of the analysis in Barro et al. (2017), the intertemporal elasticity of substitution  $\theta$  is set at 1. In this case, agents consume a constant share of their wealth, and equation (8) is replaced by

$$c_{it} = \frac{\rho + \nu}{1 + \rho}.$$

Using this relationship for  $i = 1, 2$ , and the market clearing conditions (10), (11) and (12), we obtain

$$p_t = \frac{1 - \nu}{\rho + \nu}.$$

The utility function (7) is replaced by

$$\begin{aligned} \ln u_{it} = & \frac{\rho + \nu}{1 + \rho} \ln c_{it} + \frac{1 - \nu}{1 + \rho} \ln(1 - c_{it}) \\ & + \frac{1 - \nu}{1 + \rho} \frac{1}{1 - \gamma_i} \ln \left[ \mathbb{E}_t (R_{i,t+1} u_{it+1})^{1 - \gamma_i} \right]. \end{aligned} \quad (14)$$

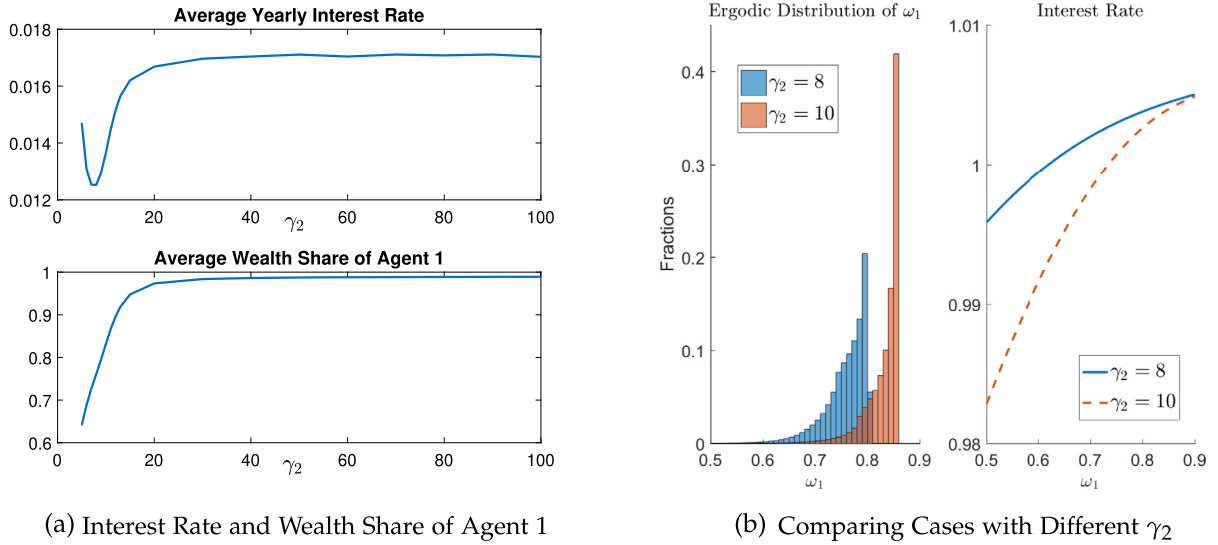
To solve the model, we use the endogenous state variable  $\omega_{1t}$  - wealth share of agents 1. This is a state variable with implicit law of motion given by (13), which correspond to equations (3) in the general framework. In each iteration and collocation point, the system of equations to be solved consists of five unknowns -  $\{x_{1t}, x_{2t}, R_t^f, \omega_{1t+1}(z_{t+1})\}$  (two values for  $\omega_{t+1}$  for two future realizations) - and five equations (9) for  $i = 1, 2$ , the market clearing condition for bond (12) and the consistency equations (13).

Since the growth shock is i.i.d.,  $\omega_1$  is the only state variable. The policy functions and stationary distributions of  $\omega_1$  are given in Fig. 4.

In Table 2 of Barro et al. (2017), the values for risk aversion parameters  $\gamma_1$  and  $\gamma_2$  are calibrated to target an average annual interest rate  $\bar{R}^f = 1.01$ . The implicit reasoning is that, for each  $\gamma_1$ ,  $\bar{R}^f$  is decreasing in  $\gamma_2$ , and there exists a value of  $\gamma_2$  such that  $\bar{R}^f = 1.01$ . Table 2 of their paper presents  $\gamma_2$  as a function of  $\gamma_1$  following this procedure. However, when  $\gamma_1 = 3.1$ , the authors set  $\gamma_2 = 10$  while acknowledging that their numerical solutions in this region were insufficiently accurate.

Using our toolbox, we can solve this problem for a wider range of  $\gamma_2$ . For smaller values of  $\gamma_2$  our results coincide with the results in Table 2 of Barro et al. (2017). But for larger values our results differ from theirs. In Fig. 5(a), we plot  $\bar{R}^f$  corresponding to different values of  $\gamma_2$  up to 100. In particular, we find that  $\bar{R}^f$  is a non-monotone function of  $\gamma_2$ . In addition,  $\bar{R}^f = 1.01$  cannot be reached when  $\gamma_1 = 3.1$  since  $\bar{R}^f$  is increasing in  $\gamma_2$  when  $\gamma_2$  is greater than 8.

The non-monotonicity is driven by two opposing forces. First, as  $\gamma_2$  becomes larger, agents 2 become more risk-averse, and demand more of the safe asset (bond). This higher demand pushes down  $\bar{R}^f$ . Second, an increase in  $\gamma_2$  also leads agent 1 to borrow more and become more leveraged. Since the return to equity is significantly higher than bond's return, the wealth share of agents 1,  $\omega_1$  tends to be larger. Larger  $\omega_1$  leads to more relative supply of safe asset and pushes up  $\bar{R}^f$ .



**Fig. 5.** Interest rate with different  $\gamma_2$ . Note: The figure is generated using the baseline parameters in Barro et al. (2017). In particular, we fix  $\gamma_1 = 3.1$  and change the value of  $\gamma_2$  to generate the results. In Figure (a), we plot the average interest rate and wealth share of agent 1 as we vary  $\gamma_2$ . In Figure (b), we compare the policy functions of  $R^f$  and ergodic distributions when  $\gamma_2 = 8$  and 10. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Whether  $\bar{R}^f$  decreases or increases in  $\gamma_2$  depends on which force dominates. Fig. 5 shows that when  $\gamma_2$  is below 8, the first force dominates and  $\bar{R}^f$  is decreasing in  $\gamma_2$  as implicitly assumed in Barro et al. (2017). However, when  $\gamma_2$  is greater than 8, the second force dominates and  $\bar{R}^f$  is increasing in  $\gamma_2$ . When  $\gamma_2$  is greater than 20,  $\bar{R}^f$  is not responsive to  $\gamma_2$  since the distribution of  $\omega_1$  becomes almost degenerated at its upper limit. See Fig. 5(b) for a comparison of two cases:  $\gamma_2 = 8$  versus  $\gamma_2 = 10$ .

#### 4.2.3. Mapping into the general setup

For the model in Barro et al. (2017) described above, the correspondence with our general setup of the toolbox is  $z = (g)$ ,  $s = (\omega_1)$ , and  $x = (c_1, c_2, x_1, x_2, R^f, K_1, b_1, p)$ .

#### 4.3. Asset pricing with heterogeneous EIS by Guvenen (2009)

Guvenen (2009) constructs a two-agent model to explain several salient features of asset pricing moments, such as a high equity risk premium, a low and relatively smooth interest rate, and countercyclical movements in the equity risk premium and Sharpe ratio. Two key ingredients of his model are limited stock market participation and heterogeneity in the elasticity of intertemporal substitution in consumption (EIS).

The solution algorithm in Guvenen (2009) is very different from ours. His is based on the algorithm in Krusell and Smith (1998): starting from conjectured laws of motion for state variables and pricing functions, he solves the agents' Bellman equation and the agents' policy functions using standard value function iterations. Then, he uses these policy functions and temporary market clearing conditions to update the laws of motion and pricing functions. He keeps iterating until the new functions are close enough to the conjectured functions.

Our algorithm recognizes that because the agents' optimization problems are *globally* concave problems, the first-order conditions are sufficient for optimality (without having to solve the agents' Bellman equation). Therefore, we can directly use policy function iterations to solve jointly for agents' optimization problems and market clearing conditions. With this algorithm, our toolbox solves the baseline model in Guvenen (2009) in less than 3 minutes on a regular laptop (about 2.9 minutes on a MacBook Air 2.2 GHz Intel Core i7) while it takes about 9 hours 15 minutes (on a MacPro Workstation with 8-core 3.2 GHz Xeon processors) using the original algorithm in Guvenen (2009) as reported in Appendix A in his paper.

##### 4.3.1. Model description

There are two types of infinitely-lived agents: stockholders ( $h$ ) with measure  $\mu$ , and non-stockholders ( $n$ ) with measure  $1 - \mu$ . Agents have Epstein-Zin utility functions

$$U_{i,t} = \left\{ (1 - \beta) c_{i,t}^{1-\rho^i} + \beta \left[ \mathbb{E}_t \left( U_{i,t+1}^{1-\alpha} \right) \right]^{\frac{1-\rho^i}{1-\alpha}} \right\}^{1/(1-\rho^i)}, \quad (15)$$

for  $i = h, n$ . Most importantly,  $\rho^h < \rho^n$ , i.e., the non-stockholders have lower EIS, which is inversely proportional to  $\rho^i$ , and thus, they have higher desire for consumption smoothness. Each agent has one unit of labor endowment.

Stockholders can trade stock  $s_t$  and bond  $b_{h,t}$  at prices  $P_t^s$  and  $P_t^f$ , respectively. Their budget constraint is

$$c_{h,t} + P_t^f b_{h,t+1} + P_t^s s_{t+1} \leq b_{h,t} + s_t (P_t^s + D_t) + W_t,$$

where  $W_t$  is the labor income, and the borrowing constraint is  $b_{h,t+1} \geq -\underline{B}$ . In the calibration  $\underline{B}$  is set at six times of the average monthly wage rate. Non-stockholders are subject to the same constraints. In addition, they are restricted from trading stocks.

A representative firm produces the consumption good using capital  $K_t$  and labor  $L_t$  based on a Cobb-Douglas production function:

$$Y_t = Z_t K_t^\theta L_t^{1-\theta},$$

and the technology evolves according to an AR(1) process:

$$\ln Z_{t+1} = \phi \ln Z_t + \varepsilon_{t+1}, \quad \varepsilon \stackrel{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2).$$

The firm maximizes its value  $P_t^s$  expressed as the sum of its future dividends  $\{D_{t+j}\}_{j=1}^\infty$  discounted by the shareholders' marginal rate of substitution process:

$$P_t^s = \max_{\{l_{t+j}, L_{t+j}\}} \mathbb{E}_t \left[ \sum_{j=1}^{\infty} \beta^j \frac{\Lambda_{h,t+j}}{\Lambda_{h,t}} D_{t+j} \right]. \quad (16)$$

The firm accumulates capital subject to a concave adjustment cost function in investment:

$$K_{t+1} = (1 - \delta) K_t + \Phi \left( \frac{I_t}{K_t} \right) K_t, \quad (17)$$

with the functional form for  $\Phi$  from Jermann (1998). Each period, the firm sells one-period bonds at price  $P_t^f$ . The bond supply is constant and equals to a fraction  $\chi$  of its average capital stock  $\bar{K}$ . Thus, dividend  $D_t$  can be written as

$$D_t = Z_t K_t^\theta L_t^{1-\theta} - W_t L_t - I_t - (1 - P_t^f) \chi \bar{K}.$$

A sequential competitive equilibrium is given by sequences of allocations

$$\{c_{i,t}, b_{i,t+1}, s_{t+1}, I_t, K_{t+1}, L_t\}$$

$i = h, n$  and prices  $\{P_t^s, P_t^f, W_t\}$  such that (i) given the price sequences,  $\{c_{i,t}, b_{i,t+1}, s_{t+1}\}$   $i = h, n$  solve the stockholders' and non-stockholders' optimization problems; (ii) given the wage sequence  $\{W_t\}$  and the law of motion for capital (17),  $\{L_t, I_t\}$  are optimal for the representative firm; and (iii) all markets clear:

$$\mu b_{h,t+1} + (1 - \mu) b_{n,t+1} = \chi \bar{K}, \quad (18)$$

$$\mu s_{t+1} = 1, \quad (19)$$

$$L_t = 1,$$

$$\mu c_{h,t} + (1 - \mu) c_{n,t} + I_t = Y_t.$$

#### 4.3.2. Computation

We use  $\{K_t, B_t^n, Z_t\}$  as the aggregate state variables, where  $B_t^n = (1 - \mu) b_{n,t}$  is the total bond holding by the non-stockholders. The optimization problems of the households and the representative firm are globally concave maximization problems, so the first-order conditions are necessary and sufficient for optimality. With this observation and the aforementioned state variables, the competitive equilibrium in this model can be represented by a system of short-run equilibrium conditions (1) required by the general framework. This system consists of 8 unknowns:  $\{c_{h,t}, c_{n,t}, I_t, B_{t+1}^n, \lambda_{h,t}, \lambda_{n,t}, P_t^s, P_t^f\}$ , and 8 equations:

1. Euler equations for bond holding:

$$P_t^f = \beta (1 + \lambda_{i,t}) \mathbb{E}_t \left( \frac{\Lambda_{i,t+1}}{\Lambda_{i,t}} \right), \quad \forall i = h, n.$$

2. Euler equations for the stockholders' demand of equity:

$$P_t^s = \beta \mathbb{E}_t \left[ \frac{\Lambda_{h,t+1}}{\Lambda_{h,t}} (P_{t+1}^s + D_{t+1}) \right].$$

3. Complementary-slackness conditions:

$$\lambda_{i,t} (b_{i,t+1} + \underline{B}) = 0, \lambda_{i,t} \geq 0, \forall i = h, n.$$

4. The budget constraints (imposing  $s_{t+1} = 1/\mu$ ):

$$c_{h,t} + P_t^f b_{h,t+1} + \frac{P_t^s}{\mu} = P_t^s + D_t + \frac{\chi \bar{K} - B_t^n}{\mu} + W_t,$$

$$c_{n,t} + P_t^f b_{n,t+1} = \frac{B_t^n}{1 - \mu} + W_t.$$

5. Firm's optimal capital accumulation  $K_{t+1}$ :

$$q_t = \beta \mathbb{E}_t \left\{ \frac{\Lambda_{h,t+1}}{\Lambda_{h,t}} \left[ \theta Z_t K_t^{\theta-1} - \frac{I_{t+1}}{K_{t+1}} + q_{t+1} \left( 1 - \delta + \Phi \left( \frac{I_{t+1}}{K_{t+1}} \right) \right) \right] \right\}, \quad (20)$$

in which capital price  $q_t$  is the Lagrangian multiplier on the capital formation (17) and satisfies

$$q_t \Phi' \left( \frac{I_t}{K_t} \right) = 1. \quad (21)$$

The auxiliary variables can be determined by the utility function (15), market clearing conditions, (17) and the following two equations:

$$W_t = (1 - \theta) Z_t \left( \frac{K_t}{L_t} \right)^\theta,$$

and

$$\beta \frac{\Lambda_{i,t+1}}{\Lambda_{i,t}} = \beta^{\frac{1-\alpha}{1-\rho^i}} \left( \frac{c_{i,t+1}}{c_{i,t}} \right)^{-\rho^i} \left[ \frac{\frac{U_{i,t+1}}{c_{i,t}}}{\left[ \left( \frac{U_{i,t}}{c_{i,t}} \right)^{1-\rho^i} - (1-\beta) \right]^{1/(1-\rho^i)}} \right]^{\rho^i - \alpha}.$$

Having represented the equilibrium in the required form (1), we can then use the toolbox to solve for a recursive equilibrium. In period  $t$ , the 6 future variables,  $c_{h,t+1}$ ,  $c_{n,t+1}$ ,  $P_{t+1}^s + D_{t+1}$ ,  $I_{t+1}/K_{t+1}$ ,  $U_{h,t+1}$ , and  $U_{n,t+1}$  are functions of  $\{K_{t+1}, B_{t+1}^n, Z_{t+1}\}$  and are solved from the previous iteration. Similar to Guvenen (2009), the initial guesses for these functions are obtained by solving a version of the model with no leverage ( $\chi = 0$ ,  $\underline{B} = 0$ ).<sup>31</sup>

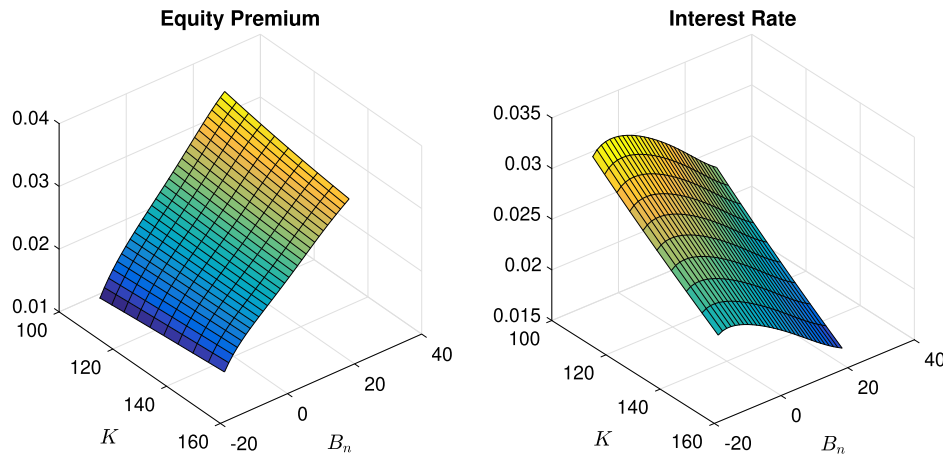
In Fig. 6, we plot the annual equity premium and interest rate as functions of  $\{K, B^n\}$  by fixing  $Z_t = 1$ . The equity premium is increasing in  $B_n$  and interest rate is decreasing in  $B_n$ . This is because stockholders, having higher EIS and tolerating more non-smooth consumption, insure non-stockholders against aggregate shocks. As  $B_n$  increases, stockholders become less wealthy, relative to non-stockholders, so they demand a higher premium to hold equity and to bear more volatility in consumption. At the same time, non-stockholders' aggregate precautionary saving increases and it pushes down interest rate. Fig. 7 plots the ergodic distributions of capital and the financial wealth share of stockholders.

#### 4.3.3. Mapping into the general setup

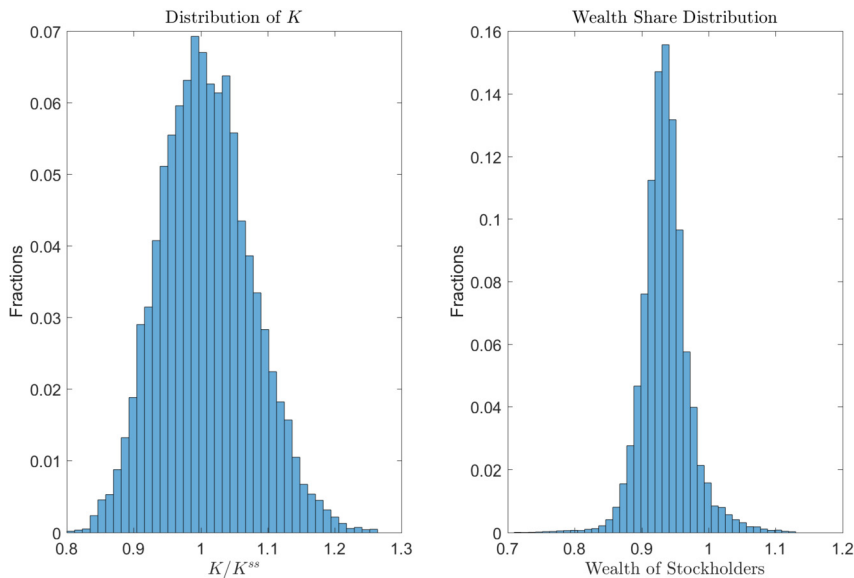
For the model in Guvenen (2009) described above, the correspondence with our general setup of the toolbox is  $z = (Z)$ ,  $s = (K, B^n)$ , and  $x = (c_h, c_n, I, B^{n'}, \lambda_h, \lambda_n, P^s, P^f, q, U_h, U_n)$ .

<sup>31</sup> It is relatively simple to implement this algorithm in the toolbox. Users can solve the no-leverage version first and after convergence, use its policy functions as the initial guesses for the full model. The toolbox allows users to override parameters and to provide their own initial guess functions using the "WarmUp" option. Thus they do not need to write separate codes for different model versions. See the codes on the toolbox's website for detail. Furthermore, the functions provided can be defined on different grid points from the state variables, which offers users much flexibility. For example, a user can solve a model with coarse grids for speed first and then use its converged policy functions as the initial guess for the same model with finer grids.





**Fig. 6.** Asset pricing policy functions in Guvenen (2009). Note: The figure plots the annual equity premium and interest rate as functions of  $\{K, B^n\}$ . We use the same parameter values as in Table 1 of Guvenen (2009), and set  $Z_t = 1$ .



**Fig. 7.** Ergodic distributions of capital and wealth share. Note: The Ergodic Distributions are generated by simulation. We use the same parameter values as in Table 1 of Guvenen (2009).

#### 4.4. Sudden stops in an open economy by Bianchi (2011)

Bianchi (2011) studies an incomplete-markets open economy model that can generate competitive equilibria featuring sudden stop episodes, mimicking those experienced by many emerging economies. A sudden stop episode features a large output drop and current account reversals, which are at odds with the prediction of a standard incomplete-markets model with precautionary saving motives. A key feature for the model in Bianchi (2011) is to introduce feedback of the price of non-tradable goods to the borrowing constraint: a negative external shock that lowers the equilibrium price of non-tradable goods tightens the borrowing constraint and forces reducing the consumption of tradable goods, which further lowers the price of non-tradable goods. The competitive equilibrium is inefficient since agents do not take into account the effects of non-tradable price on the borrowing constraint in the event of a sudden stop crisis. This leads to ex-ante over-borrowing and calls for policy interventions.

The borrowing constraint is occasionally binding in the equilibrium's ergodic set, and the equilibrium policy and state transition functions are highly non-linear when the borrowing constraint binds. Therefore, a global and non-linear solution is essential to capture the model's rich dynamics. We now describe how this class of models can be robustly and efficiently solved by the toolbox, using the exact model and calibration in Bianchi (2011) as an example.

To compute the competitive equilibrium, Bianchi (2011) uses a policy function iteration algorithm. His algorithm treats cases with binding or non-binding constraint separately, while the toolbox uses the Lagrange multiplier on the constraint

and the complementary slackness condition to write these cases using the same system of equations. This seemingly minor detail is important in allowing the model to be written and solved in the same general framework as other models. We also illustrate the use of the adaptive grid method that automatically puts more points in regions of the state space with more nonlinearity because of binding borrowing constraint.

#### 4.4.1. Model description

In a small-open economy, the representative consumer derives utility from the consumption of tradable goods  $c_t^T$  and of non-tradable goods  $c_t^N$  according to

$$\mathbb{E} \left[ \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right]$$

with the composite consumption

$$c_t = A \left( c_t^T, c_t^N \right) \equiv [\omega (c_t^T)^{-\eta} + (1-\omega) (c_t^N)^{-\eta}]^{-\frac{1}{\eta}},$$

where  $\omega \in (0, 1)$  and  $\eta > -1$  are parameters.  $\beta \in (0, 1)$  is the discount factor, and  $\sigma$  is the coefficient of relative risk-aversion.  $\mathbb{E}$  is the expectation operator to integrate shocks below.

Borrowing is via a state non-contingent bond in tradable goods at a constant world interest  $r$ . The endowments of tradable goods  $y_t^T$  and non-tradable goods  $y_t^N$  follow exogenous stochastic processes. The consumer faces the following sequential budget constraint:

$$b_{t+1} + c_t^T + p_t^N c_t^N = b_t(1+r) + y_t^T + p_t^N y_t^N,$$

where  $b_{t+1}$  is the bond-holding determined at period  $t$ . Tradable good is the numeraire, and  $p_t^N$  is the equilibrium price of non-tradable goods, taken as given by consumers.

A key feature of the model is that borrowing is subject to a borrowing constraint tied to the non-tradable good price as follows:

$$b_{t+1} \geq -(\kappa^N p_t^N y_t^N + \kappa^T y_t^T),$$

which says that the borrowing cannot exceed the sum of a fraction  $\kappa^N$  of the value of non-tradable goods, plus a fraction  $\kappa^T$  of the value of tradable goods, with parameters  $\kappa^N, \kappa^T > 0$ , determining the collateralizability of the non-tradable and tradable endowments, respectively.

A sequential competitive equilibrium corresponds to stochastic sequences

$$\{b_{t+1}, c_t^T, c_t^N, c_t, p_t^N\}_{t=0}^{\infty}$$

such that  $\{b_{t+1}, c_t^T, c_t^N\}$  solve the households optimization problem and markets clear:

$$\begin{aligned} c_t^N &= y_t^N, \\ c_t^T &= y_t^T + b_t(1+r) - b_{t+1}. \end{aligned}$$

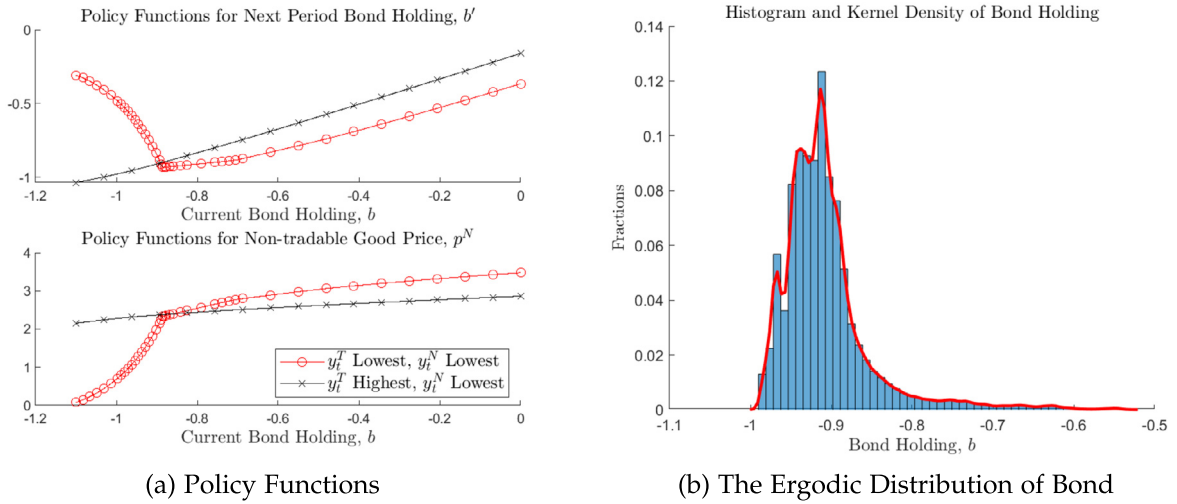
Because the households' maximization problem is a globally concave problem, the first-order conditions are necessary and sufficient for optimality: there exist stochastic processes for the Lagrange multipliers,  $\{\mu_t, \lambda_t\}$ , such that, together with  $\{b_{t+1}, c_t^T, c_t^N\}$ , the following conditions are satisfied:

$$\begin{aligned} p_t^N &= \left( \frac{1-\omega}{\omega} \right) \left( \frac{c_t^T}{c_t^N} \right)^{\eta+1}, \\ \lambda_t &= \beta(1+r) \mathbb{E}_t \lambda_{t+1} + \mu_t, \\ \mu_t [b_{t+1} + (\kappa^N p_t^N y_t^N + \kappa^T y_t^T)] &= 0, \mu_t \geq 0, \\ b_{t+1} + c_t^T + p_t^N c_t^N &= b_t(1+r) + y_t^T + p_t^N y_t^N, \end{aligned}$$

where

$$\lambda_t = c_t^{-\sigma} \frac{\partial A(c_t^T, c_t^N)}{\partial c_t^T} = c_t^{-\sigma} [\omega (c_t^T)^{-\eta} + (1-\omega) (c_t^N)^{-\eta}]^{-\frac{1}{\eta}-1} \omega [c_t^T]^{-\eta-1}.$$

With these observations, the equilibrium in this economy can be represented in the form (1) required to apply the toolbox.



**Fig. 8.** Ergodic distribution and policy functions of Bianchi (2011). Note: The policy functions are for exogenous states fixing  $y_t^N$  to be the lowest of the 4 realizations, and  $y_t^T$  to be the highest or lowest of the 4 realizations respectively. The markers indicate the grid points automatically generated by the adaptive-grid method. The histogram is based on 100 sample paths of 1000-period simulations, burning the first 500 periods of each path.

#### 4.4.2. Computation

For computation, we use the exact parameters as in the benchmark calibration in Bianchi (2011). The equilibrium can be input into the toolbox by discretizing the exogenous endowments process  $y_t^N$  and  $y_t^T$ . Following the parameterization and discretization used by Bianchi (2011), we discretize the joint process of  $(y_t^N, y_t^T)$  into 16 states. The natural endogenous state variable of the economy is  $b_t$ .

Like previous examples, a time step of policy iterations is to solve the equilibrium system defined above, for each collocation point of exogenous and endogenous states, taking the state transition function implicitly defined in  $\lambda_{t+1}(y_{t+1}^N, y_{t+1}^T, b_{t+1})$  as given. After each time step,  $\lambda_t(y_t^N, y_t^T, b_t)$  is compared with  $\lambda_{t+1}(y_{t+1}^N, y_{t+1}^T, b_{t+1})$  to check for convergence under certain criteria.

While it is possible to specify an exogenous discrete grid for  $b_t$ , since the model is highly non-linear, we illustrate the use of function approximations with the adaptive grid method provided by the toolbox, which automatically places more points in the region of the state space that features a high degree of non-linearity.<sup>32</sup> The equilibrium policy functions for  $p_t^N$  and  $b_{t+1}$ , and the ergodic distribution of  $b_t$  are presented in Fig. 8.

As shown in the left panel, the policy functions are highly nonlinear: when the borrowing constraint binds, the price of non-tradable goods declines sharply in the level of existing borrowing; future borrowing declines, instead of increasing, as the economy goes further in debt, implying current account reversals. If the borrowing constraint does not bind, then the price movement is much milder as we vary the level of existing debt, and current account reversals do not happen. The right panel displays the ergodic distribution of bond holdings, showing that non-linear regions do exist in the equilibrium ergodic set and thus cannot be ignored. But due to precautionary motives, the frequency of the economy being in these regions cannot be determined ex-ante, highlighting the necessity of using a global solution method.

The markers on the policy functions indicate the grid points automatically placed by the adaptive grid method and show that the method adds more points to the state space where the policy and state transition functions become highly nonlinear. Importantly, the method accommodates the possibility that these highly nonlinear regions differ across exogenous states, as shown in the figure. This result illustrates the effectiveness of the adaptive grid method for this class of models, as these highly nonlinear regions of state space cannot be determined ex-ante. Without the method, it would require very dense exogenous grids or painstaking manual configurations.

#### 4.4.3. Mapping into the general setup

For the model in Bianchi (2011) described above, the correspondence with our general setup of the toolbox is  $z = (y^T, y^N)$ ,  $s = (b)$ , and  $x = (b', c^T, c^N, c, \mu, \lambda, p^N)$ .

#### 4.5. Macroeconomic implications of COVID-19 by Guerrieri et al. (2022)

In this timely and important contribution, Guerrieri et al. (2022) analyze the effects of large supply shocks such as shutdowns, layoffs, and firm exits due to COVID-19. The authors show that in a two-sector model, these supply shocks can

<sup>32</sup> As described in the user manual on the toolbox's website, users only need to specify one option in the toolbox to switch to the adaptive grid method. The adaptive grid method is based on Ma and Zabaraz (2009) and Brumm and Scheidegger (2017), and features sparsity for multi-dimensional problems, and it thus can accommodate models with a high-dimensional state space.

trigger changes in aggregate demand larger than the shocks themselves, leading to a decrease in the equilibrium interest rate. This is the case when the elasticity of substitution across sectors is not too large and the inter-temporal elasticity of substitution is sufficiently high.

Their model is deterministic and the supply shock is a one time, unexpected shock. They also assume the maximally tight borrowing constraints. We extend their model to allow for stochastic, recurrent shocks and more relaxed borrowing constraints. This extension can be solved relatively easily using our toolbox. This extension shows that because these supply shocks are rare, the precautionary saving motive does not significantly change the results in the original model with a one time unexpected shock. The recurrent shocks lead to a very persistent reduction in the wealth of workers more directly affected by the shocks.

#### 4.5.1. The model

We follow closely the notation in Guerrieri et al. (2022). The total population is normalized to one, with a fraction  $\phi$  of agents working in sector 1 and the remaining fraction  $1 - \phi$  of agents working in sector 2. We assume that workers are perfectly specialized in their sector. Sector 1 is the contact-intensive sector that is directly affected by the supply shock.

The labor endowment of workers in sector 2 is constant and is set to  $\bar{n}$ , while the labor endowment of workers in sector 1 follows a two-point Markov process with state in  $\{1, 2\}$ , where 1 corresponds to normal times and 2 corresponds to pandemics. During normal times, their labor endowment is  $n_{1t} = \bar{n}$ , while when a supply shock hits, their labor endowment drops to  $n_{1t} = \delta \bar{n}$  with  $\delta < 1$ . In the COVID-19 example, sector 1 is contact-intensive, and a fraction  $\delta$  of its production is shut down when the pandemic hits. On the other hand, sector 2 is unaffected. The transition matrix between these two states is

$$\begin{bmatrix} \pi_1 & 1 - \pi_1 \\ 1 - \pi_2 & \pi_2 \end{bmatrix},$$

in which  $1 - \pi_1$  is a small probability for the economy to enter the supply-driven crisis, and  $\pi_2$  is the probability for the crisis to last for one more period.

The production technology is linear in both sectors:  $Y_{jt} = N_{jt}$  for  $j = 1, 2$ . Competitive firms in each sector  $j$  hire workers at wage  $W_{jt}$  and sell their products at price  $P_{jt}$ . Prices are flexible, and given the market structure we have  $P_{jt} = W_{jt}$ . The consumer's utility function is

$$\mathbb{E}_0 \left[ \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma} \right], \quad (22)$$

in which  $C_t = (\phi^\rho c_{1t}^{1-\rho} + (1-\phi)^\rho c_{2t}^{1-\rho})^{\frac{1}{1-\rho}}$  features constant elasticity of substitution  $1/\rho$  between the two goods and constant intertemporal elasticity of substitution  $1/\sigma$ .

As in Guerrieri et al. (2022), here we set good 2 to be the numeraire, i.e.,  $P_{2t} \equiv 1$ . Workers in sector  $j$  maximize (22) subject to

$$P_{1t} c_{1t}^j + c_{2t}^j + \frac{a_{t+1}^j}{1+r_t} \leq W_{jt} n_t^j + a_t^j, \quad (23)$$

where they allocate their labor income and bond holding from the previous period,  $a_t^j$  among consumption goods produced in the two sectors and bond holding into the next period. Interest rate  $r_t$  is determined competitively.

In addition, we assume that the workers are subject to the following borrowing constraint:  $a_{t+1}^j \geq -\bar{A}$ .

Denote sector  $j$  workers' Lagrangian multiplier for the budget constraint (23) by  $\beta^t \lambda_t^j$  and the multiplier for the borrowing constraint by  $\beta^t \mu_t^j$ . The first-order conditions for the workers' optimal decision are:

$$\lambda_t^j = (c_t^j)^{\rho-\sigma} (1-\phi)^\rho (c_{2t}^j)^{-\rho},$$

$$P_{1t} = \left( \frac{c_{1t}^j / \phi}{c_{2t}^j / (1-\phi)} \right)^{-\rho}, \quad (24)$$

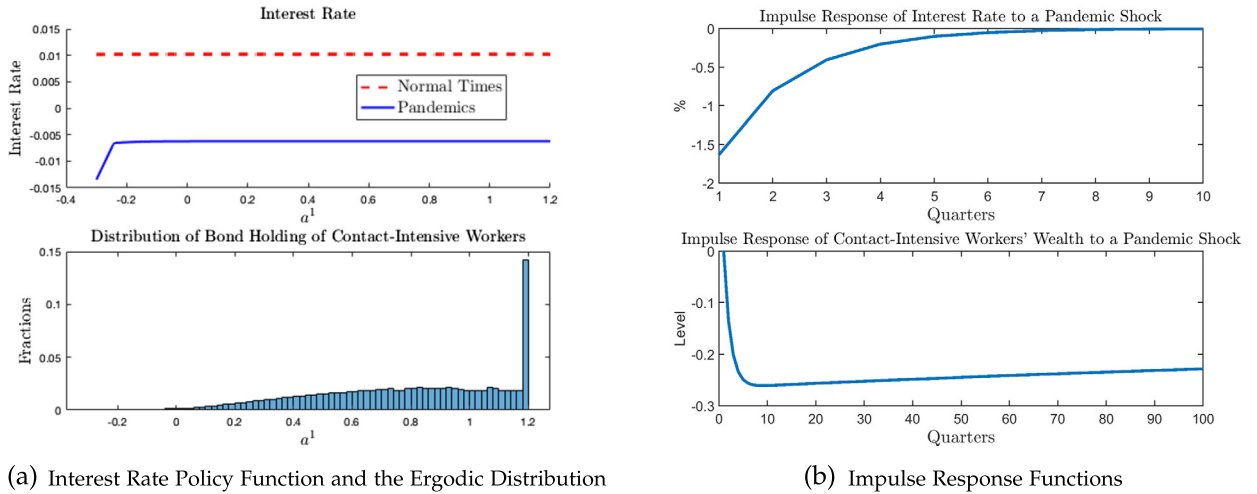
$$-\frac{\lambda_t^j}{1+r_t} + \mu_t^j + \beta \mathbb{E}_t (\lambda_{t+1}^j) = 0, \quad (25)$$

$$\mu_t^j (a_{t+1}^j + \bar{A}) = 0. \quad (26)$$

We also have the market clearing conditions for bond and consumption good 2:

$$\phi a_{t+1}^1 + (1-\phi) a_{t+1}^2 = 0,$$

$$\phi c_{2t}^1 + (1-\phi) c_{2t}^2 = (1-\phi) \bar{n}.$$



**Fig. 9.** Policy function, ergodic distribution, and IRFs. Note: We use  $\rho = 0.75 > \sigma = 0.5$  and other parameters described in the main text.

The market clearing condition for consumption good 1 is implied by Walras' law.

We use  $a_t^1$  as the endogenous state variable and look for a recursive equilibrium as a mapping from  $a_t^1$  to the allocation and prices that satisfy the first-order conditions and market clearing conditions above.

Notice that by the pricing equation (24),  $\frac{c_{1t}^1}{c_{1t}^2} = \frac{c_{2t}^1}{c_{2t}^2}$ , which means that the consumption shares of workers in sector 1 are the same between these two consumption goods. Denote the consumption share of workers in sector 1 as  $\tilde{c}_{1t}$ ; then

$$\begin{aligned} c_{1t}^1 &= \tilde{c}_{1t} n_{1t}, & c_{1t}^2 &= (1 - \tilde{c}_{1t}) \phi n_{1t} / (1 - \phi), \\ c_{2t}^1 &= \tilde{c}_{1t} (1 - \phi) \bar{n} / \phi, & c_{2t}^2 &= (1 - \tilde{c}_{1t}) \bar{n}, \end{aligned}$$

which lead to

$$C_t^1 = \frac{\tilde{c}_{1t}}{\phi} Y_t, \text{ and } C_t^2 = \frac{1 - \tilde{c}_{1t}}{1 - \phi} Y_t,$$

where  $Y_t = \left[ \phi n_{1t}^{1-\rho} + (1 - \phi) \bar{n}^{1-\rho} \right]^{\frac{1}{1-\rho}}$ , and

$$\lambda_t^1 = \left( \frac{\tilde{c}_{1t}}{\phi} Y_t \right)^{-\sigma} \left( \frac{Y_t}{\bar{n}} \right)^{\rho}, \text{ and } \lambda_t^2 = \left( \frac{1 - \tilde{c}_{1t}}{1 - \phi} Y_t \right)^{-\sigma} \left( \frac{Y_t}{\bar{n}} \right)^{\rho}.$$

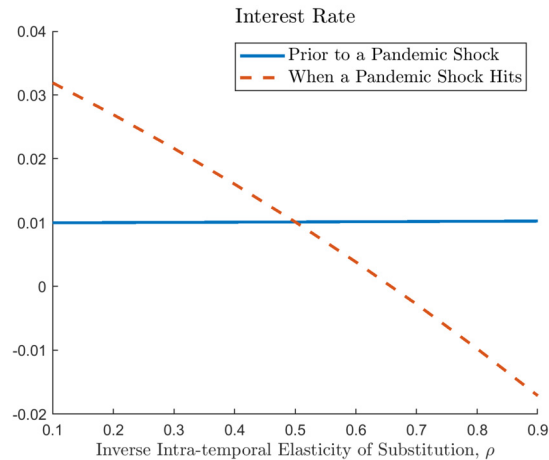
Overall, for each  $a_t^1$  (and the exogenous state of the economy), the minimal equilibrium system can be represented by five unknowns, namely,  $\{\tilde{c}_{1t}, a_{t+1}^1, \mu_t^1, \mu_t^2, r_{t+1}\}$ , and can be solved by a system of five equations, namely, the budget of workers in sector 1, equation (23), the FOC in equation (25), and the complementary-slackness condition in equation (26) for  $j = 1, 2$ .

#### 4.5.2. Calibration and results

We use quarters for model periods and standard parameters in the literature. For preferences, we use  $\beta = 0.99$  as the quarterly discount factor. The inverse inter-temporal elasticity of substitution is set at  $\sigma = 0.5$  (strictly less than 1, as required by the analytical results in Guerrieri et al. (2022) for supply shocks to trigger even larger aggregate demand responses). We vary the inverse intra-temporal elasticity of substitution  $\rho$  between 0.1 and 0.9.

For labor market parameters, we normalize  $\bar{n}$  at 1. The share of the contact-intensive sector  $\phi$  is set to 0.2. We assume that when the pandemic shocks hit, labor supply in the contact-intensive sector declines by 50% (roughly consistent with the increase in unemployment claims in the U.S. during the pandemics). We assume that the pandemics last for 2 quarters on average, equivalently,  $\pi_2 = 0.5$ .  $\pi_1$  is chosen so that the economy stays in pandemics in approximately 0.5% of the times (consistent with the historical frequency reported in Jordà et al. (2022)). Borrowing limit  $\bar{A}$  is set at 30% of the wage in normal times, a standard value in the literature.

For the benchmark results, we use  $\rho = 0.75 > \sigma = 0.5$ . The upper panel in Fig. 9(a) shows the interest rate as a function of the endogenous state variable  $a_t^1$  in normal times ( $z = 1$ ) and during pandemics ( $z = 2$ ). Interest rate is lower during pandemics, which reflects the result in Guerrieri et al. (2022) that the aggregate demand response outweighs the supply shock. In addition, the figure also shows that the effect is stronger when the net worth of workers in the contact-sensitive sector is low. The lower panel plots the ergodic distribution of bond holding of workers in sector 1. The possibility of



**Fig. 10.** Interest rate before and after pandemics. Note: We use  $\sigma = 0.5$  and other parameters described in the main text. The dashed curve corresponds to the interest rate when the shock switches from normal to pandemic, averaged in the model's ergodic set. The solid curve corresponds to the average interest rate prior to the period in which the pandemic shock hits.

pandemics leads to these workers' precautionary saving, sometimes up to the borrowing limit of workers in sector 2. However, the precautionary saving does not undo the results in Guerrieri et al. (2022). Because this is a dynamic extension of the model, we can study the dynamic responses of the economy to pandemic shocks. Fig. 9(b) shows the impulse responses of interest rate and the wealth of sector 1 workers to a pandemic shock. While interest rate reverses relatively quickly to pre-pandemic level after the shock, workers in sector 1 suffer from a persistent, long-lasting wealth loss.

To further investigate the robustness of the results in Guerrieri et al. (2022), Fig. 10 plots the average interest rate before and after the pandemic shocks hit the economy as we vary  $\rho$ . The figure shows that when  $\rho > \sigma$  (more precisely  $1/\rho < 1/\sigma$ ), interest rate drops when the pandemic shock hits, while it rises when  $\rho < \sigma$  ( $1/\rho > 1/\sigma$ ). This is exactly the result emphasized in Guerrieri et al. (2022).

#### 4.5.3. Mapping into the general setup

For the extension of the model in Guerrieri et al. (2022) described above, the correspondence with our general setup of the toolbox is  $z = (n_1)$ , and  $s = (a^1)$ , and  $x = (\tilde{c}_1, \mu^1, \mu^2, r)$ .

### 5. Heterogeneous agent models with aggregate shocks

The framework is more readily applicable to solving GDSGE models with a finite number of agents, or, more precisely, a finite number of agent types. This is because in these models, the equilibrium conditions can be represented as a system of a finite number of equations and unknowns. The solutions to these systems lie in finite-dimensional spaces. The policy and transition functions are mappings from finite-dimensional state spaces to these finite-dimensional spaces. In contrast, in fully heterogeneous agent incomplete markets models à la Krusell and Smith (1998) with both idiosyncratic and aggregate shocks, both state spaces, such as spaces of wealth distributions, and equilibrium objects, such as policy and value functions, are infinite-dimensional objects. This point is emphasized in Cao (2020).

However, Cao (2020) shows that incomplete markets models with finite agent types are useful special cases of the fully heterogeneous agent incomplete markets model in Krusell and Smith (1998). In particular, the former corresponds to the latter in which idiosyncratic shocks are perfectly persistent. We provide an explicit comparison between the two models on the toolbox's website. The dynamics of the aggregate variables in the two models are similar. This comparison suggests that, in general, the solution of the finite-agent models can be useful in understanding the properties of the fully heterogeneous agent models and can be solved at low cost using the toolbox.

In addition, the toolbox can be used to solve the agents' decision problem by observing that given a conjectured law of motion for the aggregate capital, the households' Euler equation, together with the complementary-slackness condition, is necessary and sufficient for optimality:

$$u'(c_t) = \mathbb{E} [u'(c_{t+1})(1 - \delta + r_{t+1})] + \lambda_t,$$

where  $\lambda_t$  is the Lagrangian multiplier on households' borrowing constraint and  $r_{t+1}$  is the rental rate of capital at  $t + 1$ . The toolbox can also be used to simulate the implied dynamics of the cross-sectional wealth distribution and aggregate capital. Then, with an additional fixed-point iteration on their laws of motion, which can be coded simply in MATLAB, the toolbox solution can be used to solve for a recursive equilibrium. This idea can be used to solve Krusell and Smith's baseline model in less than 100 lines of our toolbox code and 100 lines of MATLAB code. We also provide these codes, as well as the implementation for Krusell and Smith's model with heterogeneous discount factors, on the toolbox's website.



Similarly, we can use the toolbox to compute stationary recursive equilibrium in heterogeneous agent models without aggregate shocks such as Huggett (1993) and Aiyagari (1994), and transitional path equilibrium in Huggett (1997). The codes for these models are also available on the toolbox's website.

## 6. Conclusion

We provide a unified framework and a toolbox for solving DSGE models using global methods. The toolbox proves to work efficiently and robustly for a large class of highly nonlinear models, covering macro-finance, international finance, and asset pricing models. In principle, any dynamic problems characterized by systems of equations and state transition functions can readily fit in the framework, such as the decision rules in heterogeneous agent models (Huggett, 1993; Aiyagari, 1994; Krusell and Smith, 1998). The toolbox uses a policy function iteration methods and hence can be used to solve for stochastic transition paths as in Storesletten et al. (2019). Lastly, the toolbox generates Monte-Carlo simulations from model solutions. These simulations can be used for model estimation with the Generalized Method of Moments or Bayesian Estimation Methods. This is the natural next step which we leave for future developments of the toolbox.

## Data availability

All the source code will be made available on the toolbox's public Github repository

## Appendix. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.red.2023.01.001>.

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